

**Satellite Remote Sensing of Urban Heat Islands Using
MODIS Aqua MYD11A2: A Case Study on
Lagos State, Nigeria**

By

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DECLARATION

This dissertation has been composed by me. It has not been accepted in any previous application for a degree, and the work of which it is a record has been done by me. All quotations have been distinguished by quotation marks, and sources of information have been acknowledged in the text and cited in the list of references.

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ABSTRACT

This study analyses fluctuations in LST in Lagos City from 2013-2022, and the correlation between LST and NO₂ pollution. Objectives include determining UHI, mapping land-use alterations, and assessing the population at risk. The findings will offer insights into urbanisation's environmental and public health challenges delineating areas that may need interventions.

MODIS Aqua MYD11A2 and OMI AURA NO₂ tropospheric column acquired from the NASA Earthdata portal were used for this study. A fishnet 30x30 grid was created to aggregate LST and NO₂ data points, analysing their correlation using Pearson's correlation coefficient and scatter plot in SPSS. Landsat 8 satellite data was used to assess land use changes to assess changes over time from 2013 and 2022. Focal statistics were applied to fill land surface temperature data gaps due to cloud cover. GRID3 Nigeria population estimates, and the LST data created the risk map. According to the results, there were significant temperature variations during the rainy season. Maximum LST temperatures ranged from 33.92 °C in 2014 to 30.89 °C in 2022, while minimum LST temperatures recorded were 9.29 °C in 2017 and 22.13 °C in 2013. The average temperature during the rainy season ranged from 24.76 °C in 2022 to 26.55 °C in 2016, with a downward trend from 2016 to 2022. The highest temperature recorded during the rainy season generally decreased from 2014 to 2022. The LST significantly reduced during the rainy season from 2013 to 2022, indicating notable changes in temperature patterns. Maximum and mean LST showed an upward trend over the dry season from 2013 to 2022, suggesting an increase in average temperature over the period.

NO₂ concentrations during the rainy season vary yearly, with the highest levels ranging from 1.56 to 2.02 (10^{15} molecules/cm²) and the lowest from 0.47 to 0.63(10^{15} molecules/cm²). The average values also fluctuate between 0.94 to 1.12 (10^{15} molecules/cm²). Between 2013 and 2019, the mean values remained stable, with a slight decrease in 2020, followed by an increase in 2021 and 2022. Dry season NO₂ levels vary annually, with Max ranging from 2.32 to 2.69(10^{15} molecules/cm²), Min ranging from 0.72 to 1.04(10^{15} molecules/cm²), and Mean ranging from 1.41 to 1.80 (10^{15} molecules/cm²). Factors like weather influence the concentrations. From 2014 to 2016, Mean values slightly increased, followed by stable values until 2020. In 2021 there was a slight decrease, followed by a minor increase in 2022. Land-use alterations indicated a rise in developed regions by 12.63% and vegetative coverage by 5.93%, while water bodies and wetlands were reduced by approximately 1.23% and 24.26%, respectively. The population at risk of UHI was mapped, identifying eight LGAs as critical risk areas.

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Chapter 1: Introduction

1.1 Introduction to Urban Heat Islands

Conducting thorough research on surface temperatures is essential to obtain precise temperature readings in urban regions. Residents' welfare and safety largely depend on a comprehensive understanding of the energy balance in these areas. Energy transfer significantly impacts air temperature, especially in the lower parts of cities. It is crucial to comprehend these factors and their impact.(Voogt and Oke 2003). As the need for infrastructure grows, urbanisation pressures metropolitan areas, causing alterations to natural surfaces. The increase in LST, which is one of the contributors to the UHI, was caused by alterations in the terrestrial cover. (Oke, 1973; Igun, 2017).

The UHI influence temperatures increases in metropolis than rural areas and can significantly impact humans and the natural world. (Zhou et al., 2019). This phenomenon occurs in both small and large cities during the day or night. Urban areas experience two different types of heat islands: surface heat islands (SUHI) and atmospheric heat islands (AUHI). SUHIs happen due to heat absorption by urban surfaces like concrete and asphalt, while AUHIs come from human activity and pollutant concentration in the air. Both types raise city temperatures, posing health and environmental risks. SUHI is more prominent during the day and varies with seasons, while AUHI refers to the impact of heat islands in either the canopy layer (CLHI) or the boundary layer (BLHI). CLHI is measured with in-situ sensors mounted on fixed meteorological stations. BLHI is evaluated with tall towers, radiosondes, and airplanes in areas where the urban landscape does not affect the atmosphere (US EPA, 2008; Oke et al., 2017).

Climate change is influenced by two key factors: changes in land use and greenhouse gas emissions. However, distinguishing between these factors can be challenging as they contribute to higher average surface temperatures (Guha and Govil, 2021). Land cover plays a vital role in determining energy radiation and absorption, which affects land surface temperature (LST). Land coverage and surface temperature have an impact on the absorption and release of heat. (Ahmed et al., 2013).

The increase in urban populations is causing worry because of the health hazards linked to extreme heat and the UHI impact, and the potential for more frequent and prolonged heat waves caused by climate change. UHIs can worsen respiratory issues, cause heat strokes, cramps, and

dehydration, and even lead to heat-related deaths in vulnerable populations. High temperatures can cause increased energy usage, potentially leading to grid congestion and the need for controlled power cut to prevent interruptions. Fossil fuel-based energy production, used to meet increased demand during these events, produces more greenhouse gas emissions and air pollution. Moreover, thermal pollution from these emissions has a negative impact on water quality for all aquatic species. (McCartney and Mehta, 2022).

Lately, people have been increasingly concerned about the effects of urbanisation on pollutants and human well-being. Nitrogen dioxide (NO₂) is a volatile pollutant in the atmosphere that is mainly produced through vehicular movement, and it presents a significant risk to the public health. (Lamsal et al. 2013). (Sabrin, Karimi, and Nazari 2020) research showed the effects of UHIs and Ozone-PM_{2.5} pollution on urban microclimate and health. Their study revealed that many environmental and social factors contribute to increased vulnerability to extreme heat and poor air quality.

Weather changes and urbanisation on cities such as Lagos is causing various challenges. This includes elevated pressure on infrastructure, increased demands for water and energy, and a growing issue of urban poverty. Addressing this issue is crucial as it has far-reaching societal consequences (Samson et al. 2011). Ojeh et al. (2016) revealed that there could be a significant UHI impact in dry season, with temperatures exceeding 7°C at night. However, wet soils in rural areas significantly impact UHI more than wind speed during the rainy season. Through a quantitative approach, (Aderogba, 2011) revealed that reduction in land cover reduced significantly leading to changes in the microclimate influencing the intensity of UHI.

(Dissanayake et al., 2019), suggests that the growth of Lagos City and the conversion of green spaces into urban areas are significant factors contributing to SUHI. (Bassett et al., 2020) Confirmed that the proportion at which Lagos is warming has increased due to the large population already susceptible to excess heat. They also revealed that the city had caused substantial warming and atmospheric boundary layer height changes in rural areas downwind, recommending that further investigation be carried out evaluate the seasonal fluctuations in UHI in Lagos.

1.2 Aim

The work examines seasonal fluctuations in land surface temperature spanning 2013 to 2022. Furthermore, the study aims to determine the correlation between LST and NO₂ pollution and their potential impact on the populace. In addition, the research seeks to map the population prone to the UHI and NO₂ pollution in Lagos City.

1.3 Objectives

1. Evaluate UHI by analysing changes in LST throughout the seasons from 2013 to 2022.
2. Determine the relationship between the UHI intensity and air quality, specifically regarding NO₂ concentration and its potential impact on public health using Pearson correlation.
3. Assess and map the change in land use for the ten years between 2013 and 2022.
4. Assess and map the population risk.

Chapter 2: Literature Review

2.1 A Brief History of Urban Heat Islands

Urbanisation has caused a significant change in the atmosphere, particularly with the noticeable impact of the UHI. Cities have an UHI effect caused by artificial surfaces altering the environment's properties. The UHI effect can be observed in all climates and communities defined by temperature differences. Isotherm patterns of near-surface air temperatures create a heat island, which can be compared to the shape of an island. (Fernando, 2013).

(Luke Howard, 1818) conducted a comprehensive investigation of urban climate using thermometers in London and its surrounding rural areas. He discovered the UHI effect and was the first to conduct such research. Howard collected data over a period of 25 years, starting from 1806 to 1830, to identify the factors contributing to the UHI effect. Albert Kratzer's monograph "Das Stadtklima" (The Urban Climate) extensively studies urbanisation's impact on air temperature, particularly in Germany. (Albert, Vieweg and Braunschweig, 1956). Bornstein (1968) According to the investigation, the urban influence area experiences fewer and weaker surface inversions compared to the nearby non-urban strata zones. The researcher observed that higher inversion layers were more vulnerable than the rural surface inversions and frequently occurred above the city. Urban heat island intensity decreases with height, reaching zero at 310m, depending on urban-rural temperature difference.

2.2 Urban Heat Island Types

Urbanisation has led to creation of two separate heat island layers. It is essential to differentiate between these two layers, as each has distinct mechanisms. (Voogt and Oke 2003). According to this, UHI was initially grouped into two main categories (Fig 2.1) but was subsequently divided into four subcategories based on the significant classification types. (Oke, 1976; Timothy R. Oke et al., 2017). The following types of UHI are summarised below:

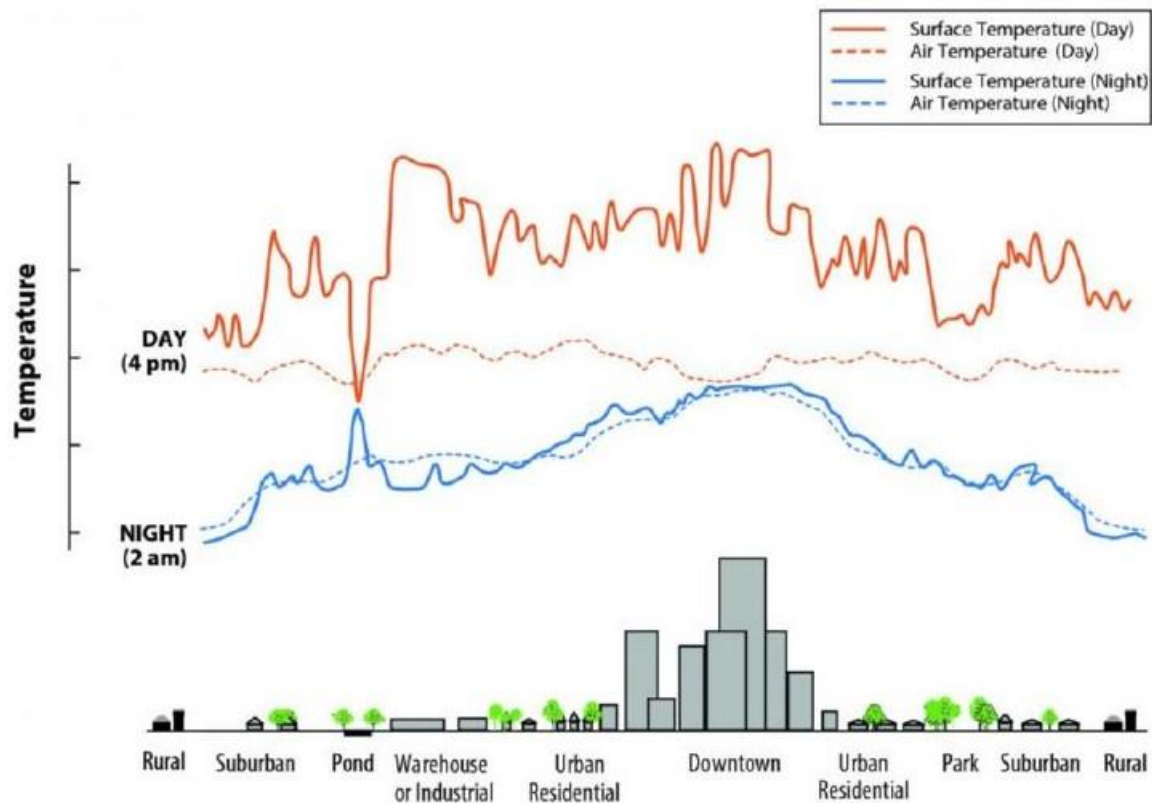


Figure 2.1: Temperature Variations on the Surface and Atmosphere(EPA 2008).

2.2.1 Atmospheric Urban Heat Islands

UHIs are pockets of hotter air in cities caused by the infrastructure retaining heat. They are most visible at night. (Oke, 1976). The summary of AUHI sub-types is summarised below:

- **Urban heat island in the canopy layer (UHI_{UCL}):** refers to the temperature contrast between the urban canopy layer (UCL) and the near-surface layer of rural areas at a similar height.
- **Boundary layer urban heat island (UHI_{UBL}):** The temperature disparity between the atmospheric boundary layer (ABL) of a rural area nearby and the layer situated amongst the top of the urban boundary layer and its top (UBL).

2.2.2 Surface Urban Heat Islands (SUHI)

Metropolitan areas experience surface heat island that occur regularly throughout the day including night time but are generally more prominent in day time. The size of surface urban heat islands is affected by seasonal variations in the sun's intensity, the amount of vegetation, and the weather. The subcategories of SUHI are described below Timothy R. Oke et al., 2017). Figure 2.2 illustrates the temperature difference between four types of UHI.

- **Subsurface urban heat island ($\text{UHI}_{\text{Subsurface}}$):** Explains the contrast in temperature patterns in rural areas and city, including the underground structures and urban soils.
- **Urban heat island on the surface ($\text{UHI}_{\text{Surface}}$):** refers to temperature variations at the point where the external atmosphere meets the built environment of a city or its counterpart in a rural area. Ideally, those interfaces would include their entire surfaces.

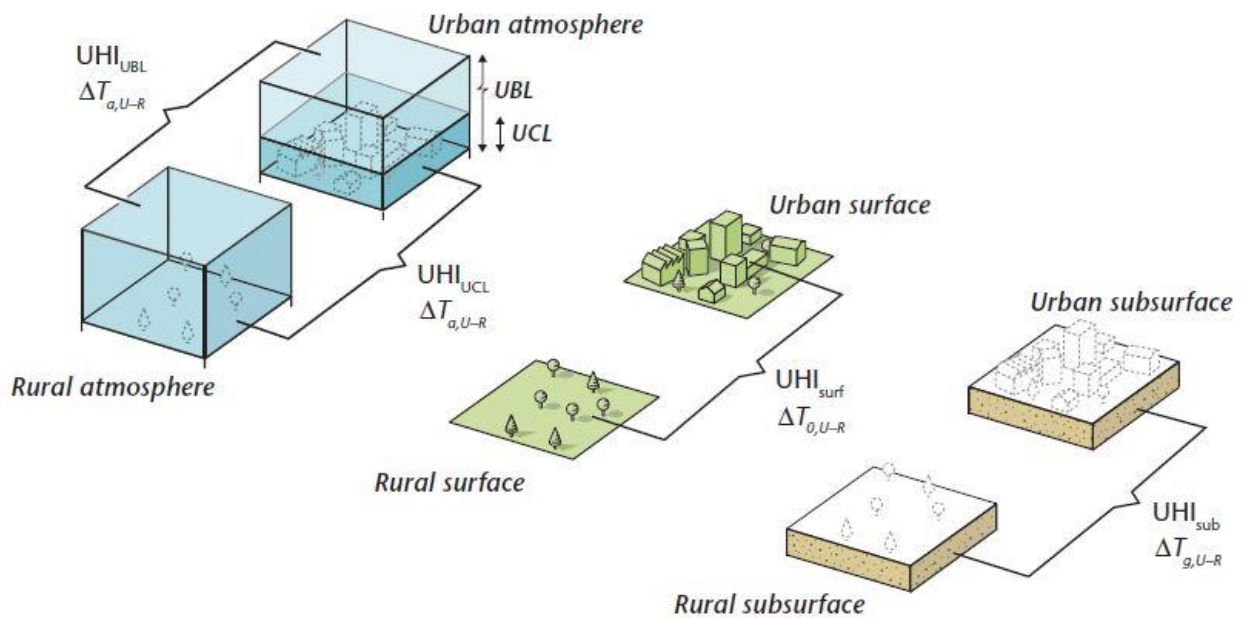


Figure 2.2: Types of UHI (Timothy R. Oke et al., 2017)

2.3 Causes of Urban Heat Islands

According to (Akbari et al. 2001; McCartney and Mehta, 2022) described the following factors influencing UHI. These factors are discussed below:

2.3.1 Albedo and Infrastructure

Urban materials' solar reflectance, thermal emissivity, and heat capacity affect the reflection, emission, and absorption of the sun's energy, which impacts the development of urban heat islands. A city's albedo might vary depending on things like surface organisation. When the surface of urban areas has a low albedo, it absorbs and retains more solar energy. This can lead to an increase in temperature within the urban environment. (Y. Huang et al. 2011).

2.3.2 Reduced Vegetation in Urban Areas

Urbanisation is causing a significant increase in dry and impervious surfaces such as roofs, walkways, roadways, etc. These areas become more dominant, greenery decreases, and concrete and asphalt structures become more prevalent. This phenomenon causes a decrease in moisture and shade, which increases in urban areas. Furthermore, the lack of water evaporation in built-up regions contributes to the rise in air and surface temperatures

2.3.3 Anthropogenic heat

The phenomenon of heat islands in the atmosphere is partly attributed to the temperature increase caused by human activities, commonly called anthropogenic heat. This factor is calculated by considering the energy consumed in various industrial processes, transportation of people and goods, use of electrical appliances, and heating and cooling systems. It is worth noting that the effects of urbanisation and infrastructure on heat generation vary, with buildings and vehicles that consume high energy levels being the main culprits.

2.3.4 Urban Geometry

Urban geometry, can influence the flow of wind, how much energy is absorbed, and the ability of surfaces to emit long waves into space.. This is because other nearby structures and surfaces can block heat flow, resulting in large thermal masses that cannot release their heat quickly. Masson (2006) referred to this phenomenon as "urban canopy," where heat is reflected by surrounding buildings but trapped by taller structures.

2.4 Influence of Urban Heat Island on Population

Extreme temperatures, heat waves, and air pollution are the primary causes of UHI, which can significantly impact human health. Residents in urban areas have a higher risk of experiencing heat-related ailments and fatalities.. This is because of several factors, such as high temperatures, inadequate nighttime cooling, low levels of surface moisture, and plants that don't experience significant temperature changes throughout the day. (Singh et al., 2020). Urbanisation has led to various health issues among city dwellers, including noise pollution, traffic congestion, inadequate housing, and air, water, and land pollution caused by industrial and transportation activities. The public health profession is deeply concerned about this trend, as it has replaced rural areas as the primary means of population distribution. (Heaviside et al., 2017).

With the rise of urbanisation come many environmental problems, including urban heat islands (UHI) and air pollution. These issues are particularly prevalent in cities, significantly impacting energy demand and consumption during warmer months. People rely on various electronic devices to maintain comfortable indoor temperatures, contributing to the upward energy demand trend. Conventional energy-producing methods are utilised to meet this demand. (EPA, 2008) In tropical and arid regions, the arrival of warmer seasons can bring about dire consequences. The sweltering heat can be unbearable for city dwellers, especially those not acclimatised to high temperatures. In such extreme conditions, these individuals may suffer from heat stress, leading to illness and even fatalities. (Nuruzzaman, 2015). Figure 2.3 illustratively shows the impacts of UHI and its implications on health.

Outdoor air pollution poses a significant risk to human health. In 2015 alone, 4.2 million fatalities occurred worldwide. (Cohen et al., 2017). It should be noted that the effects of air pollution on individuals can vary.. It varies based on several factors, including the person's health, duration and level of exposure, and the nature of pollutants present. Individuals with pre-existing medical conditions, older people, youngsters, and those from less affluent backgrounds are at the risk of pollutants because of weaker immune systems that render them more susceptible to airborne pollutants. (Singh et al., 2020).

Greenhouse gases and particulate pollutants in the air has concerning effects. This phenomenon leads to the warming of the atmosphere, resulting in lower atmospheric stability and higher pollution levels. Such a scenario presents a significant challenge for pollutants to disperse due to the heat island effect. Additionally, changes in environmental temperature profiles caused by

pollution further exacerbate the problem. The concentrations of these gases are usually highest in urban areas.(Estrada., 2017).

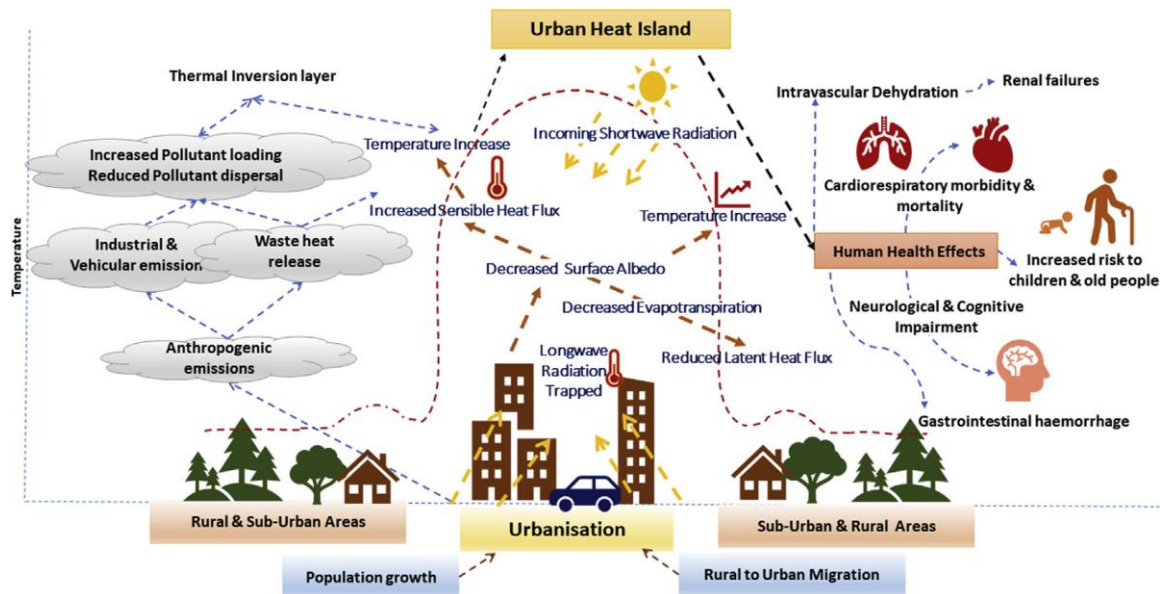


Figure 2.3: Impacts of UHI and its implications on health (Singh., 2020).

2.5 Thermal Remote Sensing of Land Surface Temperature

Thermal remote sensing focuses on collecting, analysing, and interpreting data collected mainly in the thermal infrared (TIR) part of the electromagnetic (EM) spectrum. In contrast to optical remote sensing, which measures the radiations that are "reflected" by the target under consideration, thermal remote sensing measures the radiations that are "emitted" from the surface of the target(Prakash, 2000).

An item emits electromagnetic radiation when its temperature rises above absolute zero (0 K). Every component of our environment, including plants, soil, rocks, water, and people, generates thermal infrared electromagnetic radiation with a wavelength ranging from 3.0 μm to 14 μm . Humans experience thermal energy predominantly through physical sensations. Human eyes, however, cannot discriminate between minute fluctuations in thermal infrared light transmitted from actual objects due to their primary sensitivity range of 0.4 μm to 0.7 μm . The human eye cannot see thermal infrared energy between 3.0 μm –14 μm or reflect infrared energy between 0.7 μm –3 μm (Jensen, 2014).

The top of the atmosphere (TOA) radiation that satellite TIR sensors measure the Earth's surface and atmosphere radiate. Surface emissivity (built-up regions, vegetation, bare soil), atmospheric attenuation (water vapour, aerosols), and the angle at which a satellite sensor receives the

radiation all have an impact on TOA radiances(McCartney and Mehta, 2022). For LST estimation, thermal infrared (TIR) wavelengths between 8 μ m and 15 μ m are frequently used(Tomlinson et al. 2011).

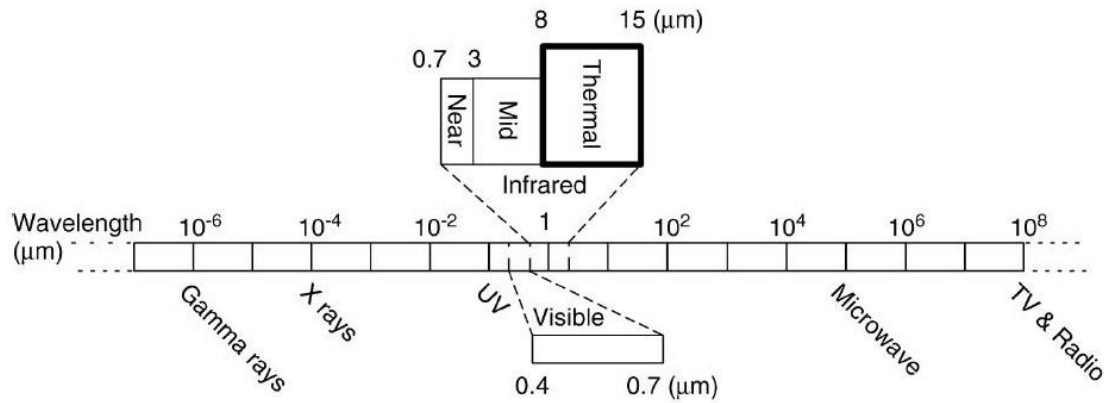


Figure 2.4: Electromagnetic Spectrum adapted from(Lillesand et al., 2015)

2.5.1 Satellites & Sensors Used for Estimating Land Surface Temperature

Thermal sensors operate in an atmospheric window between 8 μ m and 14 μ m. Heat can be sensed using RS sensors in this spectrum. A Digital Number (DN) representing the radiation is then converted into temperature data. The spectral resolution of a sensor determines its ability to collect data from the electromagnetic spectrum, including both visible and non-visible parts. The detectors are chilled to almost zero to ensure the sensors' inevitable emissions won't affect the targets' measurements and temperature records. The temperatures detected by the sensors are compared to the actual temperature of the object being measured, which is related to the absolute radiation temperature(Imhoff et al., 2010).

The satellites and sensors that are most regularly used for LST study include Landsat 8 and 9, Moderate Resolution Imaging Spectroradiometer (MODIS), Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), Advanced Very High-Resolution Radiometer (AVHRR), and Ecosystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS). The research extensively used polar and geostationary satellite data, giving the study area's temporal mapping importance(de Almeida, Teodoro, and Gonçalves, 2021). These TIR sensors can collect emissivity data on various surfaces with variable resolutions and accuracy in addition to LST readings. Urban climate and environmental studies use LST and emissivity data to analyse patterns and surface characteristics, assess UHIs, and connect LSTs with surface energy changes. This helps to identify terrain properties, patterns, and

processes.(Lo, Quattrochi, and Luvall, 1997). Table 2.1 summarises the primary satellite and sensors used to measure LST.

Table 2.1: Common Satellite and Sensors Used for LST Estimation Adapted from (McCartney and Mehta, 2022).

Satellite	Sensor	Temporal Coverage	Spectral Bands (μm)	Spatial Resolution	Temporal Resolution
Landsat 4	Thematic Mapper	7/1982 – 12/1993 3/1984 – 01/2013 4/1999 – Present	10.40 – 12.50	120m (30m resampled)	16-days
Landsat5 Landsat 7	Enhanced Thematic Mapper (ETM+)	02/2013 - Present	10.40 – 12.50	60m (30m)	
Landsat 8	Operational Land Imager (OLI) Thermal Infrared Sensor(TIRS)		10.6 -11.19 11.50 – 12.51	100m	
MODIS Terra	Advanced Spaceborne Thermal Emission and Reflection Radiometer & MODIS	12/1999 – Present	10.78 – 11.28 11.77 – 12.27	1km	12 hours
MODIS Aqua	Moderate-resolution Imaging Spectroradiometer (MODIS)	04/2002 - Present			
ASTER	Terra	03/2000 - present	10.25 – 1095 10.95 – 11.65	90 m	12 hours
NOAA Operational Series Current: NOAA 15,18,19 ESA – Metop A& B	AVHRR	1979 - Present	10.30-11.30 11.50 -12.50	1km and 4km	
International Space Station/ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS	Prototype HypsIRI Thermal Infrared Radiometer (PHyTIR)	06/2018 - Present	8.28,8.79,9.06, 9.06, 10.5, 12.05	60m Conterminous United States and critical biomes around the world.	Varies/ everyday

2.5.2 Monitoring Surface Urban Heat Islands through Satellite Remote Sensing

Urban climatology may benefit significantly from and actively seek satellite-based sensors that can give Sun-synchronous high spatial resolution thermal imagery. The UHI event unintentionally contributes to weather changes, which has significant practical implications for water and energy conservation, enhancing human comfort, lowering air pollution, and circulating local air (Roth, M et al., 1989). The integration of thermal remote sensing with urban micrometeorology is a significant advancement, creating new opportunities for monitoring UHIs and investigating their causes. UHIs have since been remotely observed via satellite and aircraft platforms to examine several urban heat island features thanks to their data (Voogt and Oke, 2003).

The first study on SUHI utilising satellite remote sensing was published by Rao in 1972. He conducted the investigations using the Television Infrared Observation Satellite (ITOS-1) environmental satellite. Noticed a difference in temperature between the land and the ocean. He concluded that on a clear night with low wind, the warmth of a city could be felt at night. (Rao, 1972). Urban heat island size correlates with population size. (Oke, 1973). He believed that if the sun is out, population logarithms and the inverse of local wind speed are correlated with heat island intensity.

Thermal fingerprints of various municipal types throughout daytime and nighttime can explain how they affect the UHI phenomenon. It was found that business, manufacturing, and service uses are in areas with higher midday irradiance than those with lower water levels, agriculture, and vegetation. Due to its exceptionally high emissivity, which shows only a very slight difference in irradiance between day and night, water stands out as the temperature is higher at night and lower at daytime (Lo et al., 1997). To comprehend the workings and reactions of the environment, it is necessary to obtain accurate readings of surface energy flow and temperature through thermal infrared (TIR) remote sensing. An illustration of this is the accurate computation of evapotranspiration and soil moisture using TIR remote sensing data. TIR remote sensing data can improve accuracy of regional daily evapotranspiration measurements. (Quattrochi and Luvall, 1999).

Satellite imaging technology like MODIS, ASTER, AVHRR, or Landsat thermal imagery is used to study properties of metropolitan city. This involves analysing surface activities, materials, and structures to determine spatial patterns of brightness or temperature.(Voogt and Oke, 2003). These TIR sensors can gather surface emissivity data with varied degrees of precision and

resolution. The landscape's characteristics, practices, and processes are described using LST and emissivity data. They are typically used to evaluate UHI, investigate LST trends and how they relate to surface properties, and establish a connection between LSTs and surface energy fluxes(Quattrochi and Luvall, 1999).

About 78% of all LST studies employ the two most popular satellite sensors, MODIS Terra and Aqua, and Landsat 5 Thematic Mapper (TM), Landsat 7 Enhanced Thematic Mapper (ETM+), and Landsat 8 Thermal Infrared Sensor (TIRS) (Zhou et al. 2019). The phenomenon of urban heat sink was studied in central Indiana using data from Landsat 5 Thematic Mapper (TM). The results showed the typical heat island effect in the metropolitan area tends to be warmer than in rural areas. This is attributed to specific surface conditions, such as dry, bare, low-density soil (Carnahan and Larson, 1990). Nichol (1994) used Landsat 5 TM data to study the microclimates of cities during the daytime, when it is most likely that there are higher levels of thermal stress on human activities, to give a more realistic representation of the possible influence on nearby air temperatures of particular surfaces in cities, his opinion was that there were greater than 3 metres of significant urbanisation characteristics or feature groups which affected nearer ground temperature. Kaplan et al., (2018) used Landsat 8 to study the impact of urban heat islands during summer. They examined how green spaces and built-up areas affect UHI. Guha et al., (2020), Observed the interaction between LST and NDVI and concluded that their relationship remains consistently inverse across all seasons.

A comparison was made between ground-based measurements of LST and MODIS Terra LST. The ground temperature measurements were more representative during the daytime than at night, Confirms MOD11A1 LST at a resolution of 1 km ground-based LST measurements has a strong correlation. (Wang et al., 2007). (Imhoff et al., 2010) reported UHI increases as the city expands and is higher in forested ecosystems during summer. This can increase energy usage for cooling, providing insights for future urban development approaches. A study analysed ten years of MODIS data to see how topography and land cover impact LST variations. The landscape affects LST differently depending on land coverage and time of day. (Stroppiana et al., 2014). Furthermore, a predictive analysis of urban heat island (UHI) studies was conducted using MODIS. The investigation revealed a strong correlation, confirming that urbanisation contributes to the UHI effect. However, other factors also play an essential role in UHI and were found to be statistically significant parameters(Jain et al, 2019). In their study, (Badugu et al., 2023) utilised the MOD11A2 LST product to examine fluctuations of LST and UHI impacts.

They validated that the innermost region of a city, with elevated concentration of constructed structures, encountered the highest average LST of 25.56°C per year.

2.5.3 Methods of Acquiring LST Data for UHI Study

The most frequently used methodologies consist of the Split-Window Algorithm (SW), Mono-Window Algorithm (MWA), Single Channel Method, MODIS Day/Night Algorithm, and Temperature Emissivity Separation Algorithm (TES). The algorithms mentioned are discussed below.

2.5.3.1 Split-Window Algorithm

The split-window algorithm (SWA) is widely used for compensating atmospheric effects. It is a direct approach algorithm that does not require solving the radiative transfer equation or inputting external atmospheric profiles. SWA uses two bands in the atmospheric window region to adjust for disturbances. These bands are typically between 10-12µm and have differential absorption characteristics. (Hulley et al. 2019). Anding et al. (1970) were the first to apply the split window technique, Which entails utilising space-based systems to scan a particular region and acquire more precise evaluations of water surface temperature. Radiometric measurements are taken in two thermal infrared wavelengths. The measurement of sea temperature remotely using thermal infrared wavelengths is based on the spectral radiance emitted by a grey body given by:

$$L_{\lambda}^{\epsilon} = \epsilon(\lambda) L_{\lambda}^b (T) \quad (1)$$

The spectral emissivity of the grey body is $\epsilon(\lambda)$ while the $L_{\lambda}^b (T)$ is the radiance emitted by a black body. This was later represented as

$$L_{\lambda}^{\epsilon} = \frac{2c^2h}{\lambda^b[(\exp hc/\lambda KT)-1]} \quad (2)$$

The variables stated here are significant factors determining the temperature of a blackbody. These include T (temperature), c (speed of light), h (Planck's constant), k (Boltzmann's constant), and λ (wavelength).

(Wan and Dozier, 1996), Proposed generalised split-window algorithm to compute LST from MODIS data. The algorithm has been successfully utilised to generate MODIS LST products and has been validated.. The expression of the algorithm for MODIS is as follows:

$$T_S = C + (A_1 + A_2 \frac{1-\epsilon}{\epsilon} + A_3 \frac{\Delta\epsilon}{\epsilon^2}) \frac{T_{31}+T_{32}}{2}$$

$$+(B_1 + B_2 \frac{1-\varepsilon}{\varepsilon} + B_3 \frac{\Delta\varepsilon}{\varepsilon^2}) \frac{T_{31}-T_{32}}{2} \quad (3)$$

The SW coefficients are represented by A, B, and C. The average and the variance of the emissivities for bands 31 and 32 are denoted by ε and $\Delta\varepsilon$, respectively.

2.5.3.2 MODIS Day/Night Algorithm

(Wan and Li, 1997) developed MODIS day/night algorithm that uses principles of physics to measure surface temperatures and emissivities. This algorithm relies on data from seven thermal infrared bands collected day and night. A regression method was used to solve 14 nonlinear equations. The algorithm was tested in different atmospheric conditions, temperatures, and emissivities, while considering MODIS instrument specifications.

2.5.3.3 Temperature Emissivity Separation (TES) Algorithm

Isolating temperature and emissivity are complex due to six unknowns and five measurements. ASTER's TES uses a hybrid approach that blends two algorithms. It assesses temperature and emissivity using the normalised emissivity method and normalises emissivities by their average value. The TES method predicts minimum emissivity and improves precision using empirical data and iterative processes. It requires land-leaving radiances and downwelling sky irradiance. TES can recover temperatures with a deviation of ± 1.5 K and emissivities with a variation of ± 0.015 . Limitations include reliance on empirical relationships, The accuracy of ASTER's measurements and atmospheric correction, as well as compensation for reflected sky irradiance, are important factors to consider..(Gillespie et al., 1999). Figure 2.5 shows the TES iterative approach.

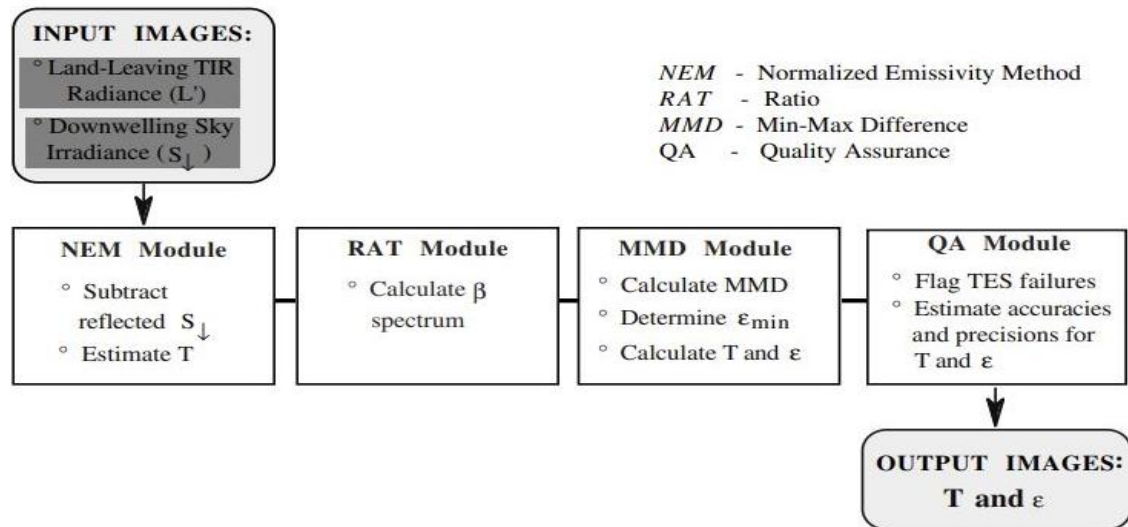


Figure 2.5: Basic Iterative Process of TES (Gillespie et al. 1999).

2.5.3.4 Single Channel Algorithm

The generalised single-channel method measures water vapour and channel wavelength for satellite thermal sensors. It's unique because it doesn't need in-situ radio soundings or mean atmospheric temperatures. It is appropriate for use different sensors with the same equation and coefficients. (Jiménez-Munoz and Sobrino, 2003).

2.5.3.5 Mono-Window Algorithm

(Qin et al., 2001) Developed a technique called the mono-window algorithm that enables the computation of land surface temperature using Landsat TM data. Although this method is dependable, radiometric calibration may still be needed in specific applications. The algorithm is based on the thermal radiance transfer equation and the linearisation of Planck's radiance function. Three critical parameters are employed in the algorithm: emissivity, transmittance, and mean atmospheric temperature, which renders it feasible to estimate LST from Landsat TM data. Furthermore, this algorithm can also be employed to derive LST from other sources of one-thermal-channel data. In 2013, the Landsat 8 satellite launched with two thermal infrared bands improved the method of obtaining land surface temperature data. The improved mono-window algorithm uses local meteorological data and land cover approaches to estimate essential parameters and has successfully retrieved LST data from Landsat 8 TIRS data.(F. Wang et al., 2015).

Chapter 3: Methodology

3.1 Introduction

This section aims to study the seasonal variation of LST and its relationship with air pollutants, with a specific focus on NO₂. To achieve this goal, a systematic workflow is presented to analyse the study area comprehensively. The study employed data from the MODIS Aqua MYD11A2 and OMI AURA NO₂ tropospheric columns acquired from the NASA Earthdata portal. A fishnet 30x30 grid was created to aggregate LST and NO₂ data points to determine their relationship by applying Pearson's correlation coefficient and scatter plot in the SPSS application. The Landsat 8 satellite data was also utilised to calculate LULC changes and establish the relationship between UHI and urbanisation for December 2013 and 2022. (See Fig 3.1).

3.2 Study Location

Lagos State is a vast urban metropolis in Nigeria's southwestern region. It covers an area of 3,577 square kilometres, approximately 903,066 acres, and accounts for 0.387% of the country's total land area. Despite its relatively small size, Lagos is home to a projected inhabitants of 27 million as of 2020, with an annual growth rate of 3.2%. This makes it home to the largest population in Nigeria and a rapidly growing city worldwide. Lagos is the centre of the nation's trade and commerce, responsible for over 70% of the country's industrial and industrial activities. Its strategic location, bustling seaports, and vibrant economy make it a prime destination for local and foreign investors. (MEPB – Lagos, 2021).

Lagos State is geographically positioned within the latitudinal range of 6° 25'N – 6° 33'N and the longitudinal span of 2° 25'E – 4° 22'E (Figure 3.1). This region is renowned for its picturesque low-lying coastal area along the expansive Atlantic Ocean. The stunning coastline features sandy beaches, wetlands, and marshes. The land slopes from inland to coast, covering the lagoon and spanning east to west. The elevation of this region varies from sea level to 25 meters above it. Additionally, the northern part of the state boasts three distinct upland zones that add to its allure, with an elevation of 70 meters above sea level (Nkeki Felix Ndidi and Ojeh Vincent Nduka, 2014).

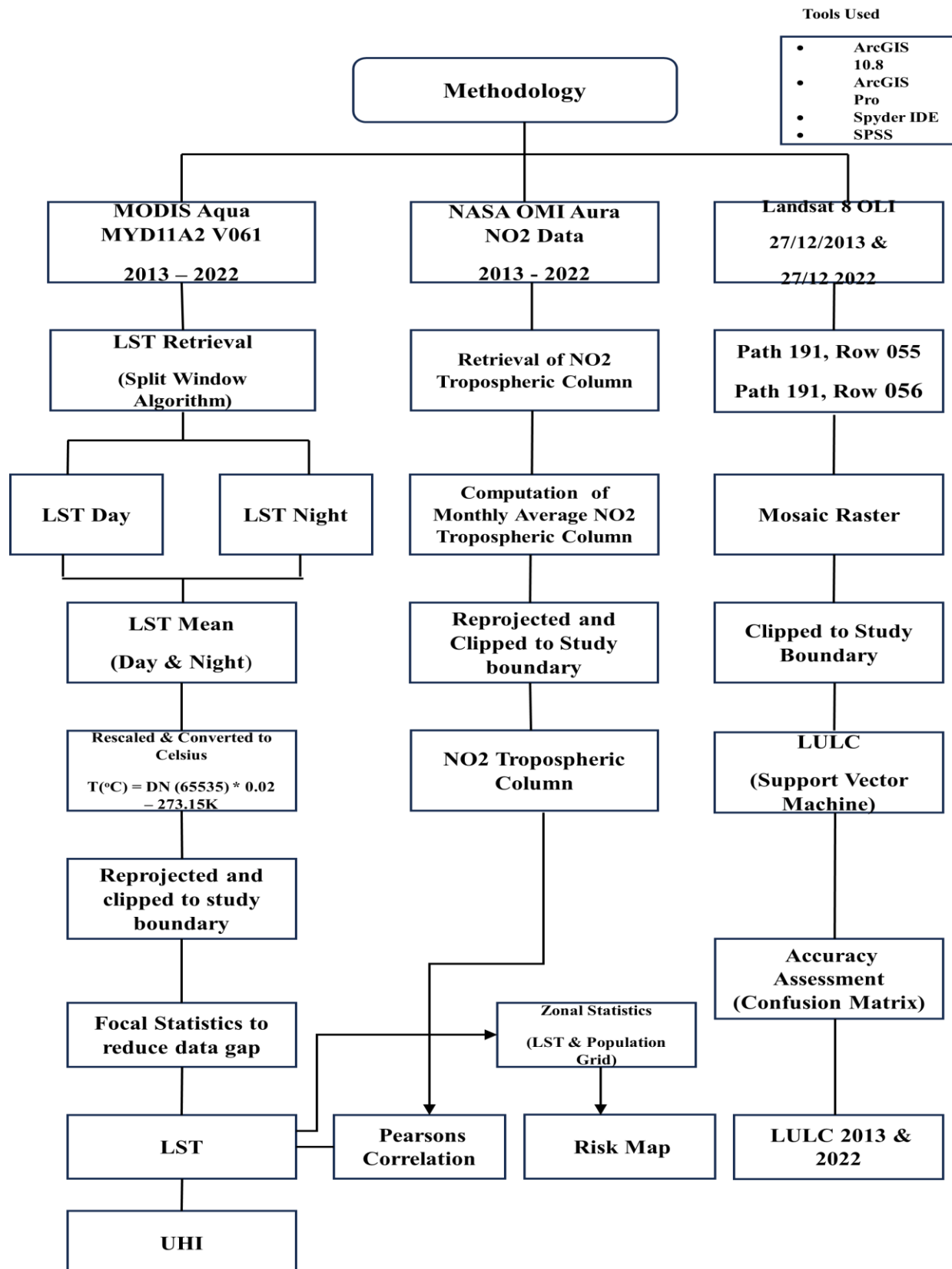


Fig 3.1: Study workflow

Lagos State's weather patterns are affected by its location near the equator and the Gulf of Guinea, resulting in a wet equatorial climate. This climate is mainly controlled by atmospheric interactions, specifically the Intertropical Convergence Zone (ITCZ). The Inter-Tropical Convergence Zone (ITCZ) heavily influences Lagos' environment. When it advances northward, it brings warm, humid air from the ocean, while the retreating ITCZ brings hot, dry air from the land. Lagos has two seasons. Rainy season from April to October, dry season from November to March. Heavy rainfall in the rainy season due to warm and humid air, less rain in the dry season due to hot and dry continental air. Despite the fluctuating weather conditions, Lagos State consistently experiences high temperatures, with an average monthly maximum temperature of approximately 30 °C. This makes Lagos warm and humid destination throughout the year(BNRCC, 2012).

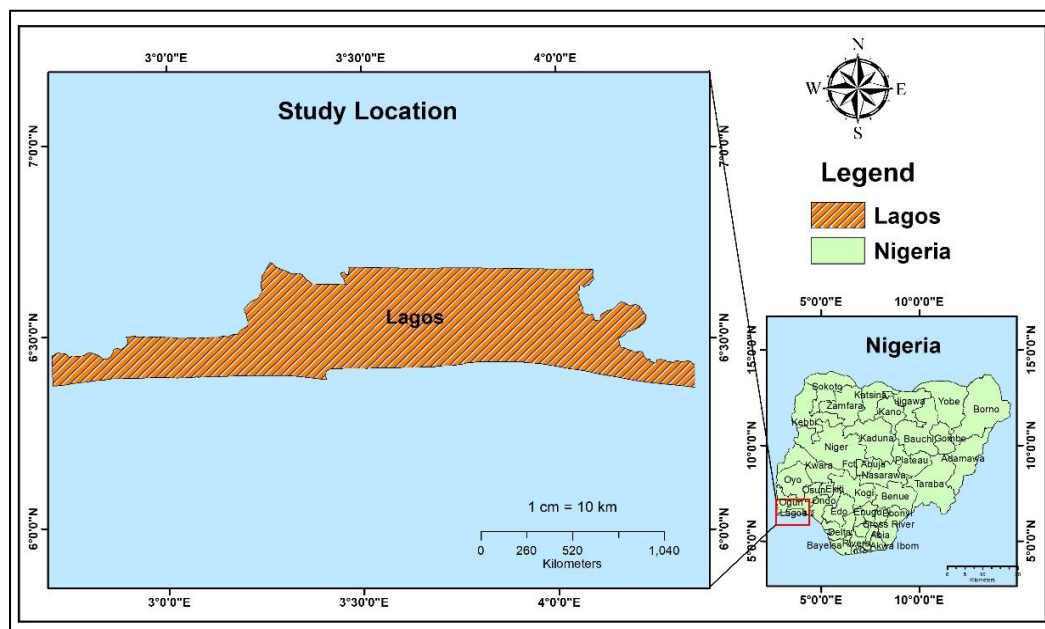


Fig 3.2: Map of Study Location

3.3 Data Preparation and Techniques

3.3.1 MODIS MYD11A2

Through the NASA Earthdata portal <https://urs.earthdata.nasa.gov/>, 552 MODIS Aqua MYD11A2 V6.1 granules were successfully obtained, covering the period from 2013 to 2022. These granules measure LST and emissivity for eight days with a 1km spatial resolution in a 1,200 x 1,200km grid. (Wang et al. 2007). The study aimed to analyse temperature variations in the urban region of Lagos by collecting data during both day and night periods from 2013 to

2022. To achieve this, the generalised split-window technique (equation 1) and an eight-day revisit interval were utilised to assess temperature changes during different seasons. Our data source was MODIS MYD11A2, with a repeated cycle time of 1330pm local time. Using a Python script within the Spyder IDE in the ArcGIS Pro environment, The land surface temperature for dry and rainy seasons was computed and processed. MODIS data were reprojected from the Sinusoidal tile grid used in creating the MYD11A2 products to WGS 1984. Next, we re-scaled it with a scale factor of 0.002, converted it to Celsius, and clipped it to the study boundaries. The process of calibrating the LST and its conversion to Celsius is expressed in Equation 4 (Wan 2019).

$$T(^{\circ}\text{C}) = \text{DN} (65535) * 0.02 - 273.15\text{K} \quad (4)$$

$T(^{\circ}\text{C})$ is the LST, and DN represents the digital number of the MODIS satellite data.

Urban Heat Islands are warmer in cities than in rural areas. A viable method of calculating the UHI involves utilising an equation within the raster calculator given in equation 5 (Chaudhary et al. 2021)

$$UHI = \frac{T_s - T_m}{SD} \quad (5)$$

The formula follows: T_s = land surface temperature, T_m = LST mean, and SD = standard deviation for the LST data.

3.3.2 OMI AURA NO2 Tropospheric Column

Since its deployment in July 2004, the Ozone Monitoring Instrument (OMI) has worked tirelessly on the NASA EOS Aura spacecraft. As a crucial part of our planet's environmental monitoring system, the OMI's primary objective is to maintain a reliable and extensive archive of total column ozone levels to measure air pollutants. It also closely tracks the concentration of other trace gases that contribute to global tropospheric pollution. The OMI is a highly advanced UV-Vis spectrometer located strategically on the polar-orbiting NASA Aura satellite. follows a sun-synchronous orbit, crossing the Equator at approximately 1345 p.m.(Levelt et al. 2018).

Nitrogen dioxide (NO_2) is a harmful air pollutant can have serious consequences on people and the environment. It can originate from various sources, including natural causes like lightning and volcanic eruption, as well as human interactions with the environment such as transportation and power generation. NO_2 can react with other atmospheric pollutants to form harmful

compounds like nitric acid (HNO₃) and ozone, leading to respiratory problems and other health issues. Satellite-based observations are used in conjunction with traditional ground-level measurements to monitor and understand NO₂ levels on a global scale accurately. These observations provide a comprehensive record of NO₂ concentrations and can help identify areas at a higher risk for harmful pollution levels. By improving the accuracy of these measurements, we can work towards reducing the effect of NO₂ . (N. Lamsal et al. 2021).

For this work, the primary dataset of the Level-3 daily global gridded Nitrogen Dioxide Product (OMNO2d) was used. This data provides detailed pixel-level information that is of good quality and is then aggregated and averaged into 0.25x0.25-degree global grids. The data contains information on the Total Column NO₂ and Total Tropospheric Column NO₂ daily, but only in cases where the cloud fraction is less than 30%. This applies to all atmospheric and sky conditions, ensuring reliable and accurate data (Krotkov et al. 2016). Three hundred sixty-five global gridded granules were acquired with a resolution of 0.25x0.25 degrees. The NO₂ tropospheric column with a 30% cloud screen was utilised to estimate the seasonal average. Subsequently, the data was reprojected and resampled to match the MODIS MYD11A2 cell values, using bilinear interpolation to ensure consistency in the final output.

3.3.3 Land Use Land Cover Classification

Landsat 8 data was downloaded from the USGS Earth Explorer to quantify the LULC change between December 2013 and 2022. The imagery is classed into four types: built-up area, vegetation, water bodies, and wetlands. This will estimate the percentage change in land use over ten years.

Table 3.1: Landsat 8 Data used for the LULCC

SN	Data	Sensor	Band	Path and Row	Resolution	Date
1	Landsat 8	OLI	10	Path 191, row 055	30m	2013 – 12 - 27
2	Landsat 8	OLI	10	Path 191, row 056	30m	2022 – 12 - 27

3.3.4 Focal Statistics

A significant limitation of land surface temperature data acquired via remote sensing is its inability to capture observations for regions obstructed by cloud cover (Crosson et al. 2012). Due to this reason, the Focal Statistics tool was used to compute a neighbourhood operation to fill the gaps and produce a raster output. The process involves calculating the neighbourhood statistics by summing up all the cell values within the designated neighbourhood and the processing cell. To achieve this, the conditional statement was employed in map algebra-raster calculator, using “*Con(IsNull("lstd_project"), FocalStatistics("lstd_project", NbrRectangle(5, 5, "CELL"), "MEAN"), "lstd_project")*”. Within this work's context, Two figures, namely Figures 3.2 and 3.3, illustrate the process of utilising the raster calculator alongside the map products. These figures showcase the state of the map products before and after applying the focal statistics, thereby providing a clear representation of the operations involved in this process.

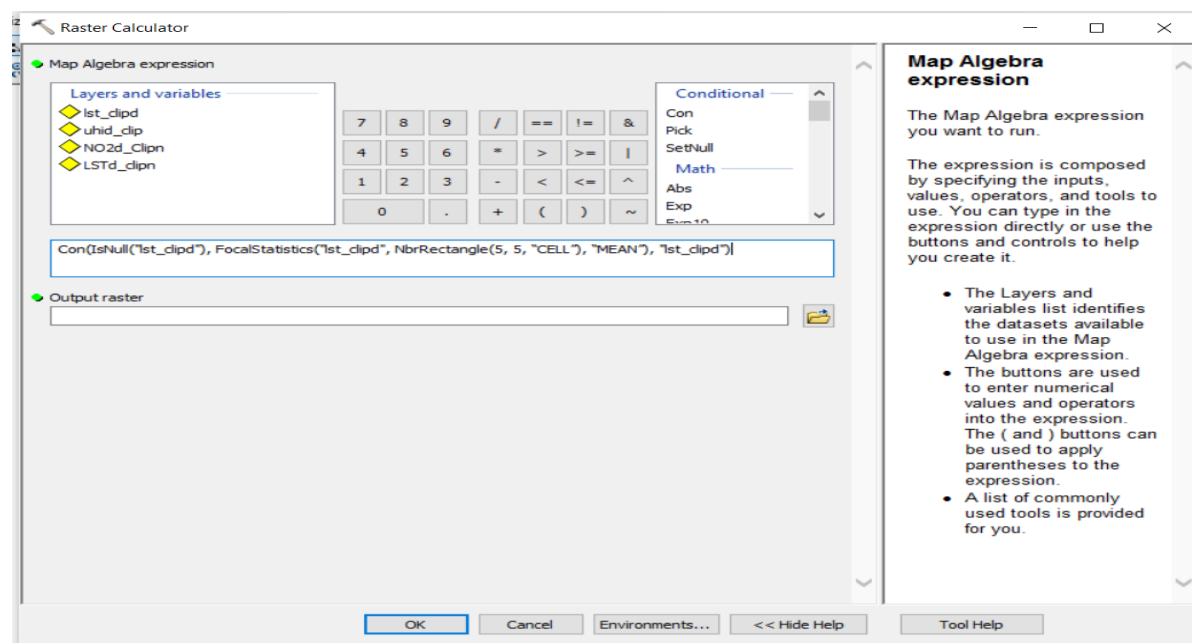


Figure 3.2: Focal Statistics Raster Calculator operations applied to fill the data gaps

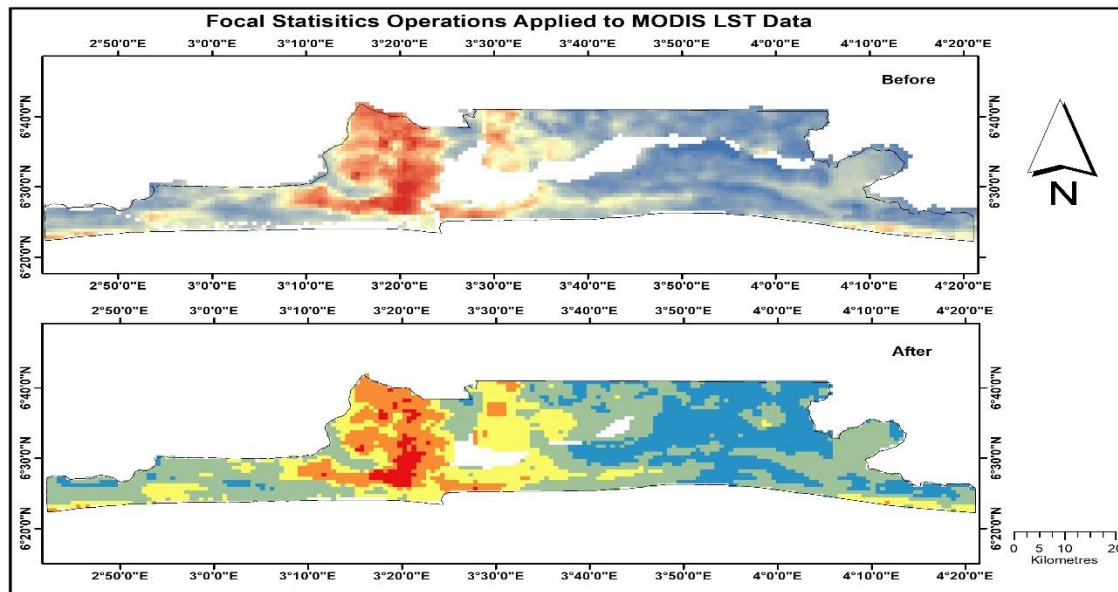


Figure 3.3: Shows the Map Product Before and After the Focal Statistics Operations

3.3.5 GRID3 Nigeria Population Estimates

The WorldPop Research Group at the University of Southampton conducted an extensive study to estimate the population of Nigeria's administrative units, focusing on levels 2 and 3. This critical research was a part of the GRID3 project, which aimed to gather accurate and reliable data to support national development efforts. The study's findings provide valuable insights into the distribution and density of Nigeria's population, which can inform policy decisions and resource allocation. The data release provides estimated population statistics for Nigeria with high spatial resolution at three arc-seconds, equivalent to approximately 100m grid cells. This data offers national coverage and estimates of populations categorised by age and gender. It includes up-to-date and more comprehensive settlement information and an alternate regional boundary definition aligned with the time the satellite imagery was processed to generate building footprints. The population estimates are only mapped in regions with predicted residential settlements, resulting in higher accuracy and completeness. (WorldPop, 2020).

3.3.6 Fishnet Toolset

To create features of sample data points for locations or areas, Fishnet is a valuable tool set. To make a fishnet, you need three types of information: the spatial extent of the fishnet, the number of rows and columns, and the rotation angle. In this study, we used the Create Fishnet tool to generate 30 x 30 grid cells of rows and columns. We then extracted aggregate data points containing a net of rectangular cells of sampled locations for correlation analysis (ESRI, 2023).

3.3.7 Pearsons Correlation Coefficient

Analysing the possible correlation between two continuous variables is a crucial statistical technique called association analysis. Pearson's correlation coefficient evaluates the intensity and direction of the linear relationship. But if the variables have a non-normal distribution, using Spearman's correlation coefficient is better. This coefficient depends on the data's rank order and calculates the dominance and trend of the monotonic association connecting the parameters. This study evaluated the association linking the land surface temperature and nitrogen dioxide using Pearson's correlation coefficient(Mukaka 2012), which was calculated in SPSS. The Pearson's correlation coefficient format is given as follows:

$$r = \frac{\Sigma(X_i - \bar{X}) * \Sigma(Y_i - \bar{Y})}{\sqrt{\Sigma(X_i - \bar{X})^2 * \Sigma(Y_i - \bar{Y})^2}} \quad (6)$$

3.3.8 Zonal Statistics

To ensure accurate zonal operations, it's essential to have a clear understanding of zones and their corresponding values. It's worth noting that a zone encompasses all areas with the same value, regardless of whether they're contiguous. Matching resolutions are highly recommended, as any differences in resolution will trigger an internal resampling process before the zonal operation (ESRI, 2023). In this study, a zonal statistics operation was carried out to map the vulnerable population at risk of UHI and concentration using the gridded population data and UHI data layers.

Chapter 4: Results

4.1 Analysis of the LST and NO₂ Tropospheric Column Concentration Variations by Season (2013-2022)

This section delves into the analysis of fluctuations in land surface temperature and NO₂ Tropospheric column concentration spanning from 2013 to 2022. At the core of the LST measurement lie the temperature readings recorded at 1330 p.m. over the MODIS Aqua MYD11A2 sensor. Pearson's correlation method present us the opportunity to calculate the correlation coefficient and the R² scatterplot to determine the magnitude and direction of direct correlation between the two variables. This allowed us to ascertain if there was a correlation between these two variables during both seasons of the period under study. To further enhance the scope of this analysis, an attempt was made to identify and map vulnerable populations at risk. By doing so, areas that need urgent intervention can be identified. This step is significant as it allows us to identify the administrative areas that may require intervention due to the impact of UHI and NO₂ Tropospheric column density on the urban population.

4.1.1 Rainy Season LST Variations (2013 – 2022)

During the rainy season, various temperature variations were observed (See Figures 4.1 – 4.10). The LST maximum temperatures varied between 33.92 °C in 2014 and 30.89 °C 2022. The LST minimum temperatures recorded were 9.29 °C in 2017 and 22.13 °C in 2013. Notably, there was an increase in the minimum temperature between 2017 and 2020, followed by a decrease in 2022. The average temperature recorded during the rainy season ranged from 24.76 °C in 2022 to 26.55 °C in 2016. The overall trend from 2016 to 2022 showed a downward trajectory, indicating a decrease in the average temperature (see Figure 4.11). Furthermore, the highest temperature recorded during the rainy season showed a general downward trend from 2014 to 2022, signifying a decrease in the most elevated temperatures. Between 2013 and 2022, there was a noticeable decrease in LST values during the rainy season. Specifically, the maximum LST decreased between 31.75 °C - 30.89 °C, while the minimum LST increased from 22.13 °C to 18.22 °C. In addition, the average LST also reduced from 25.41 °C to 24.76 °C. These observations indicate significant changes in temperature patterns during the rainy season over time, as mentioned above (See Table 4.1).

The line plots provide a detailed visual representation of the maximum, minimum, and mean land surface temperatures (LST) during specific years. The maximum LST plot depicts a remarkable decrease in the highest recorded temperatures during the rainy season. This trend is

highly encouraging, as it suggests that the environment is undergoing a cooling effect, which is critical for maintaining ecosystem balance. Conversely, the Minimum LST plot unveils temperature fluctuations, where there was a sudden increase in 2017, followed by a subsequent decline in 2022. Further analysis is required to determine the factors that may have triggered this variation. Lastly, the Mean LST plot displays a general decrease in temperature, indicating that the average temperature during the rainy season is gradually declining. It's essential to keep track of any positive environmental changes to make sure they keep getting better.

4.1.2 Dry Season LST Variations (2013 – 2022)

The maximum land surface temperature throughout the dry season fluctuated between 30.74°C in 2016 and 32.49°C in 2019. From 2013 to 2019, there was a noticeable increase in maximum LST, with the highest peak occurring in 2019. This suggests an overall warming trend during the dry season. In 2020, the maximum LST slightly decreased but considerably increased in 2021. The LST has increased overall during the specified time. The minimum LST trends during the dry season also varied from 22.65°C in 2016 to 23.97°C in 2019, similar to the maximum LST. However, The average LST varied during the dry season from 26.03°C in 2013 to 27.18°C in 2021, gradually increasing from 2013 to 2021 and peaking in 2021. The mean LST data indicates a warming trend during the dry season, similar to the maximum LST data.

The data shows that LST varies yearly, which can be influenced by climate patterns, land use changes, and local conditions. The data suggests an increasing trend in maximum and mean LST over the dry season from 2013 to 2022. Although there is variability in individual years, the general trend indicates increasing temperatures during this period. The calculated temperature change is 1.09°C, It indicates that there has been a rise in the mean temperature during the dry season between 2013 and 2022.

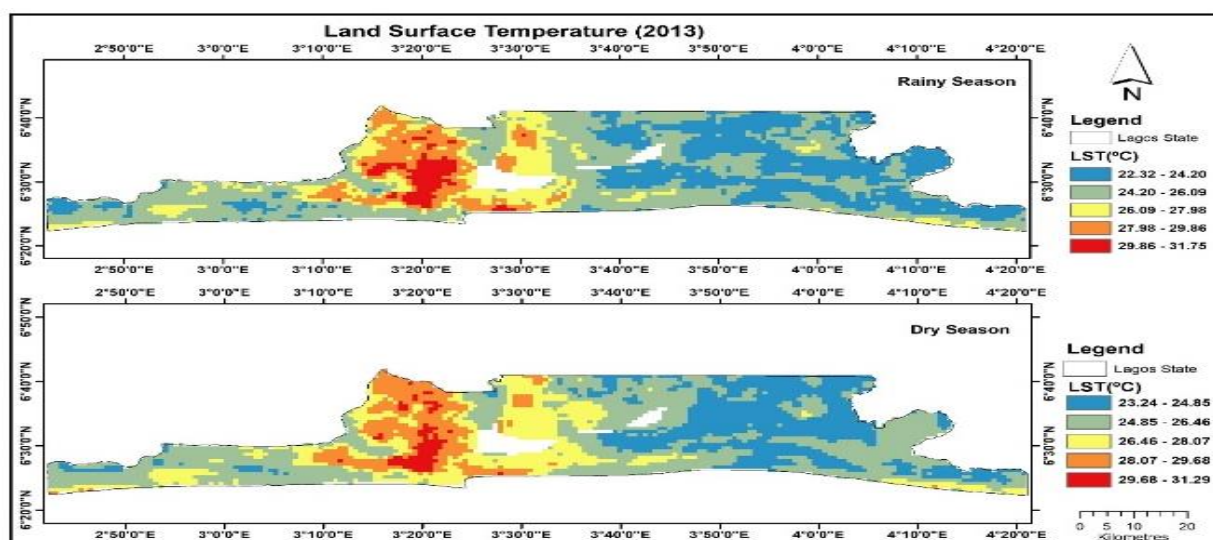


Fig 4.1: LST 2013 (Rainy and Dry Season)

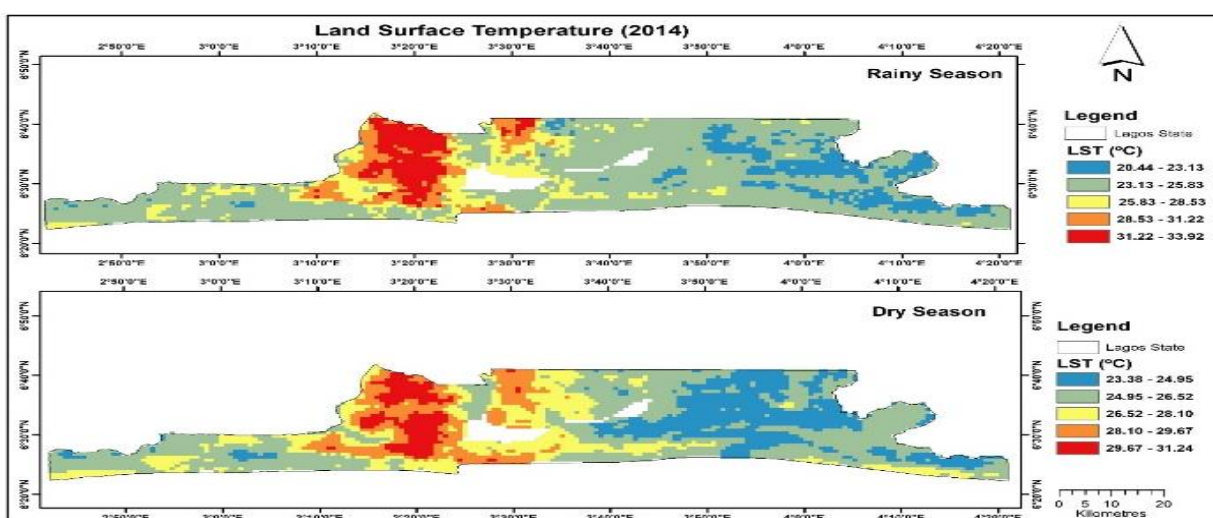


Fig 4.2: LST 2014 (Rainy and Dry Season)

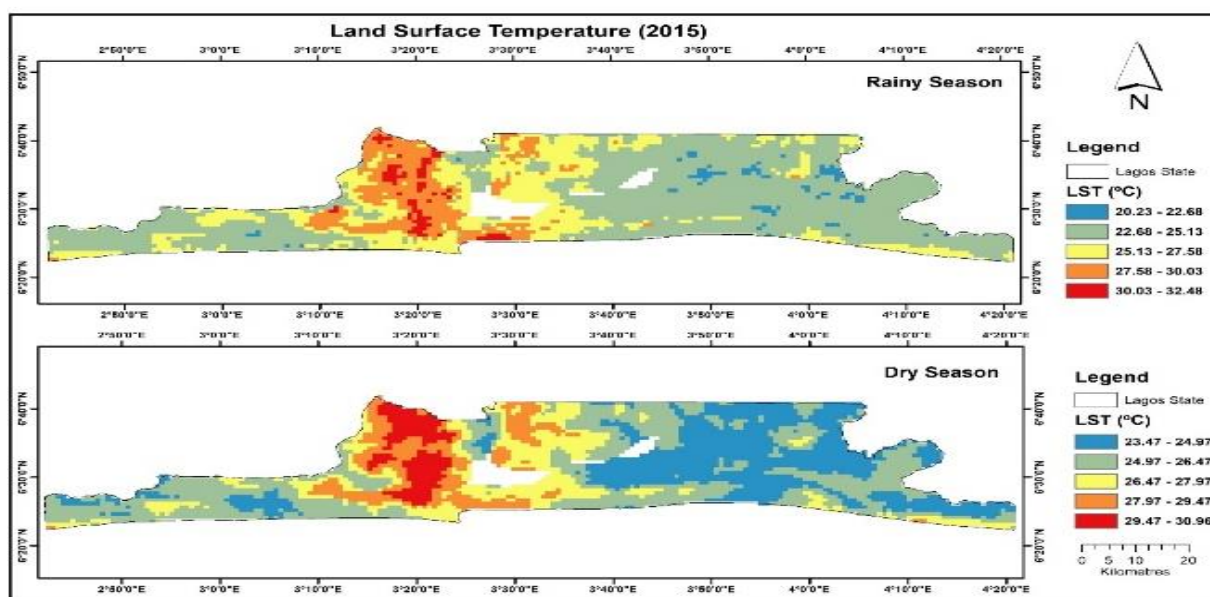


Fig 4.3: LST 2015 (Rainy and Dry Season)

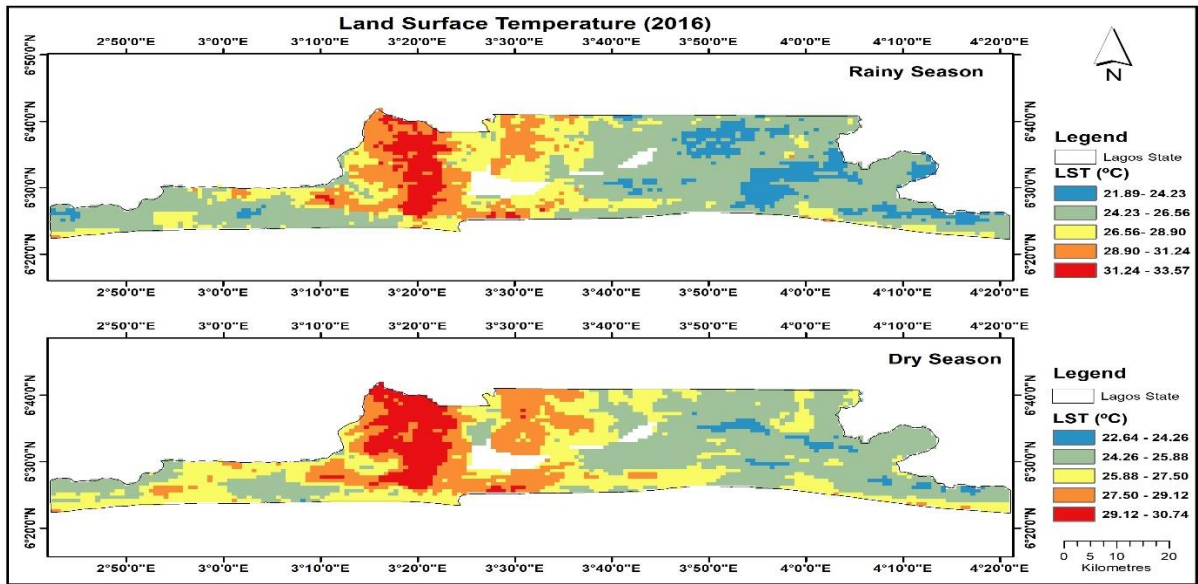


Fig 4.4: LST 2016 (Rainy and Dry Season)

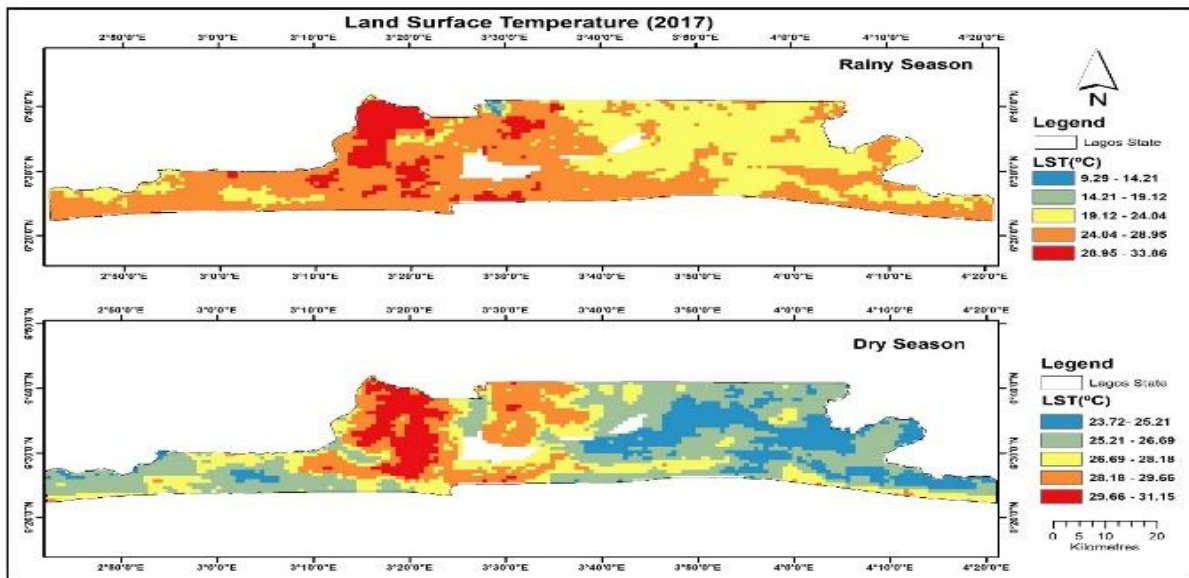


Fig 4.5: LST 2017 (Rainy and Dry Season)

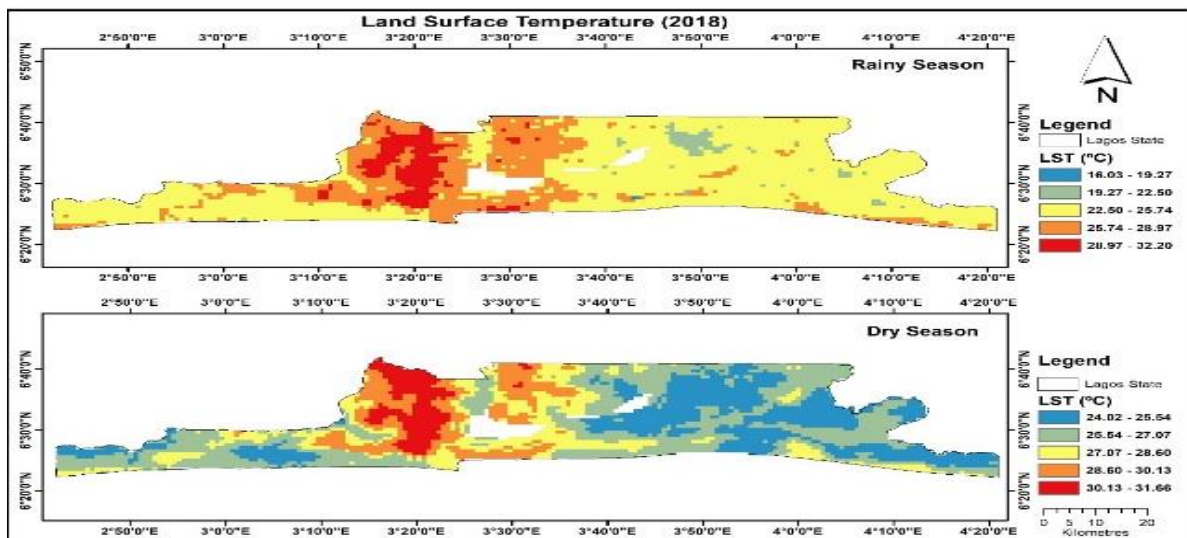


Fig 4.6: LST 2018 (Rainy and Dry Season)

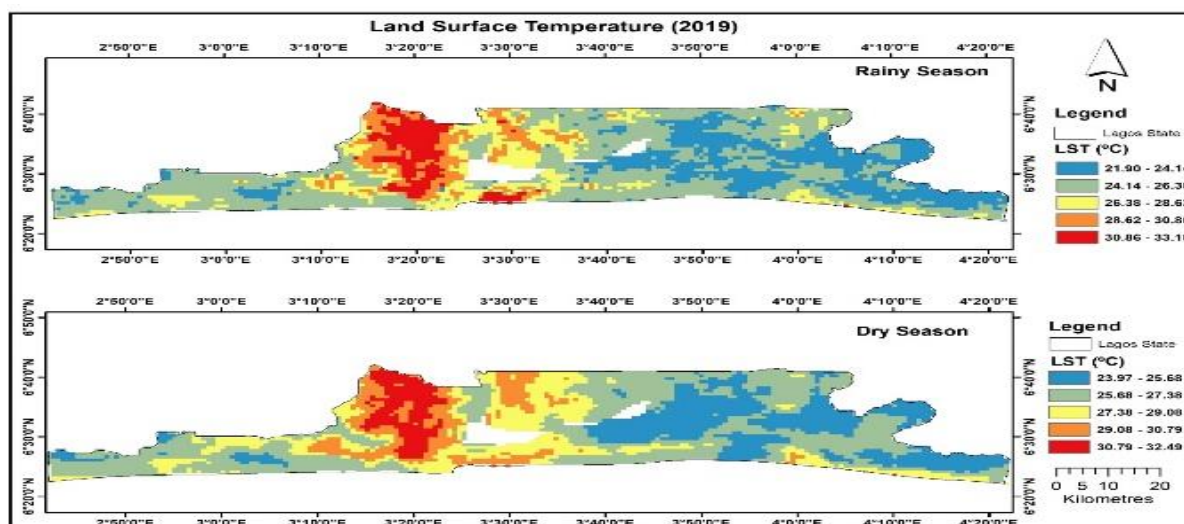


Fig 4.7: LST 2019 (Rainy and Dry Season)

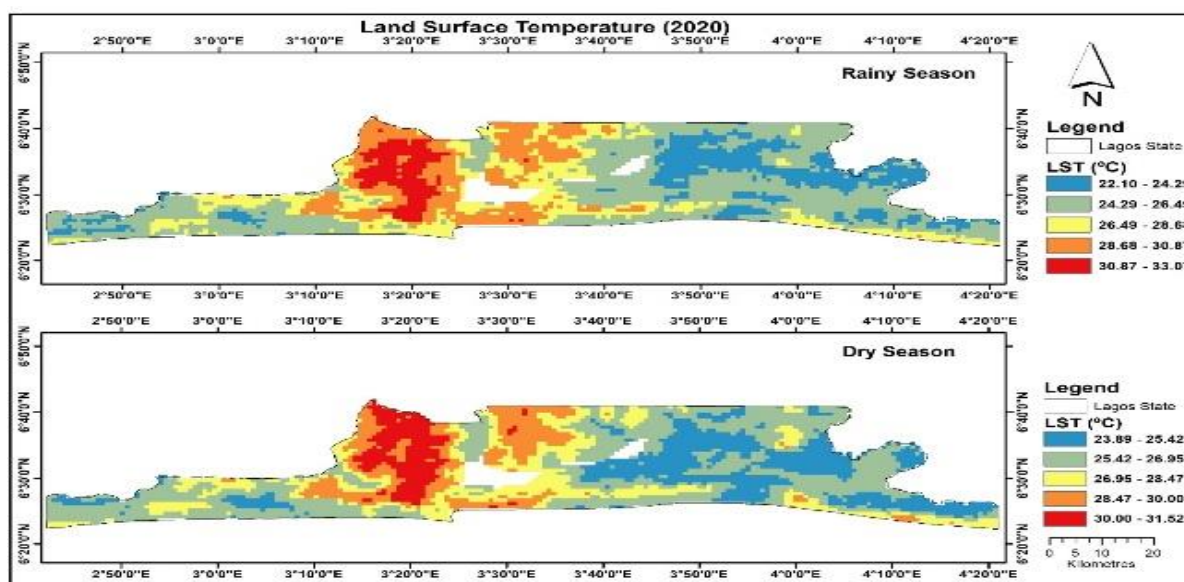


Fig 4.8: LST 2020 (Rainy and Dry Season)

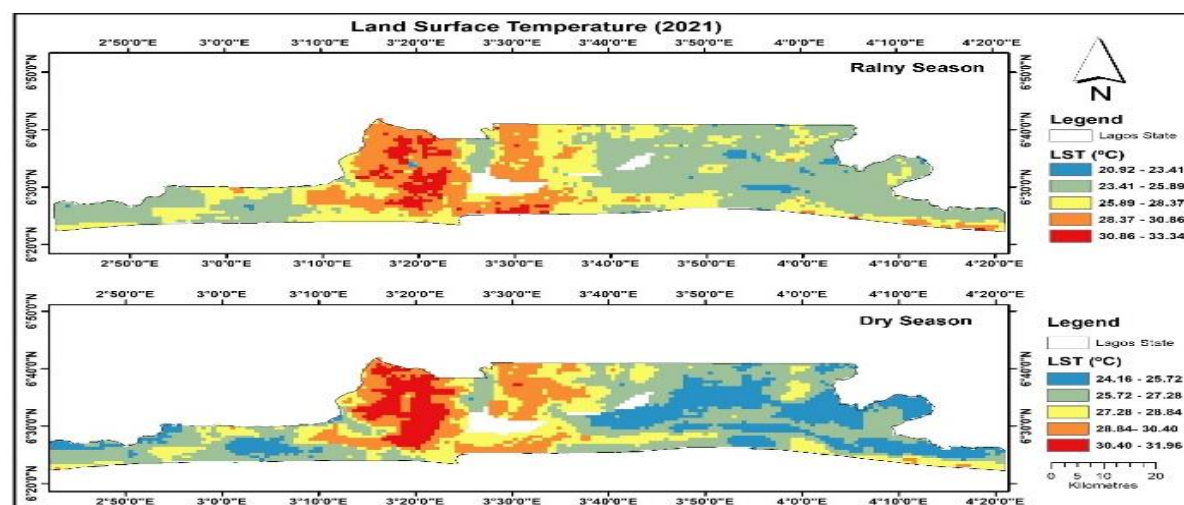


Fig 4.9: LST 2021 (Rainy and Dry Season)

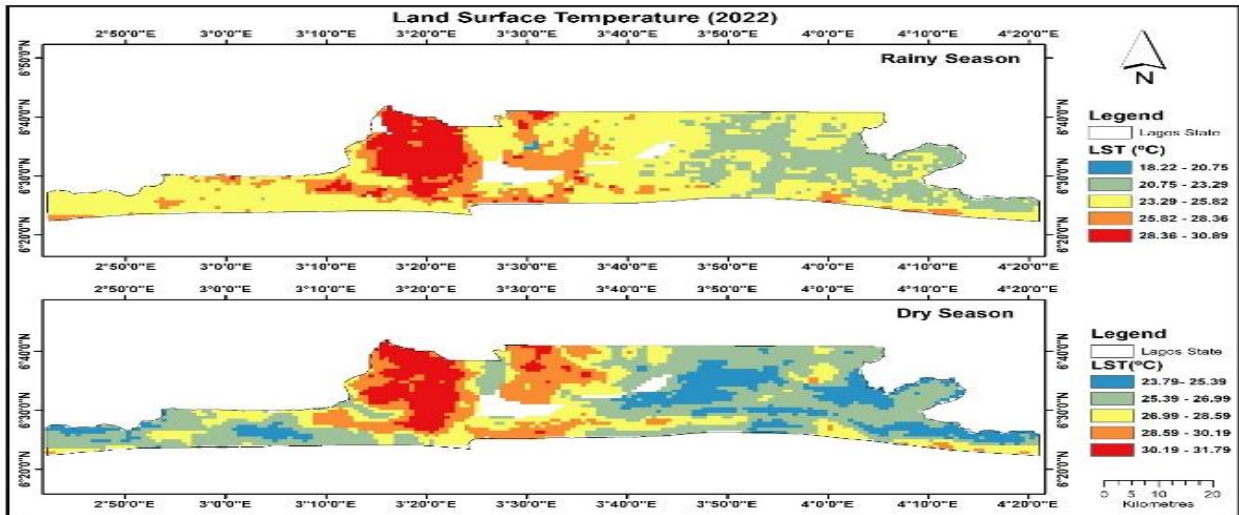


Fig 4.10: LST 2022 (Rainy and Dry Season)

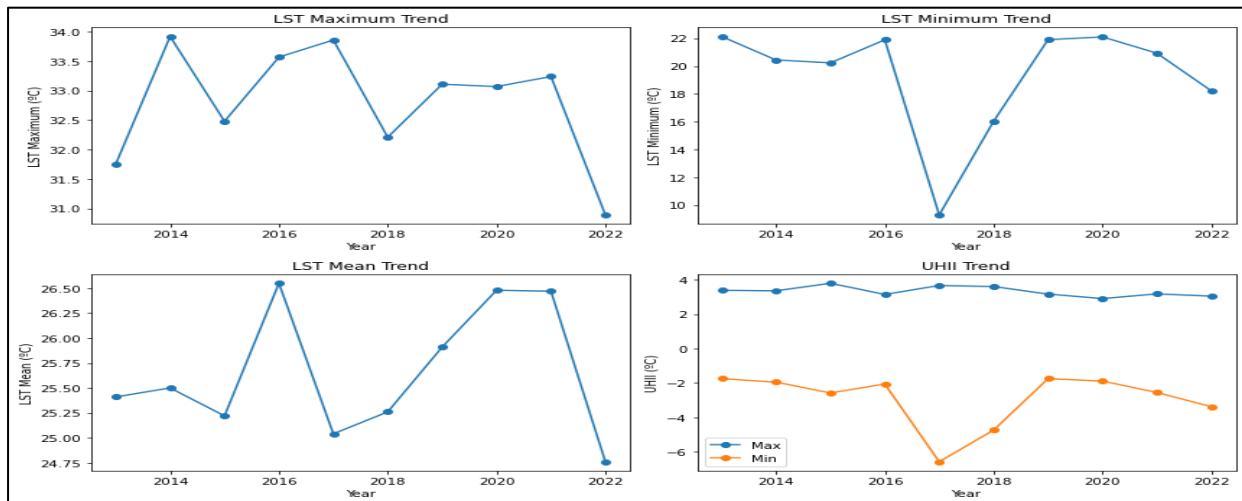


Figure 4.11: Rainy Season LST Trends (2013 – 2022)

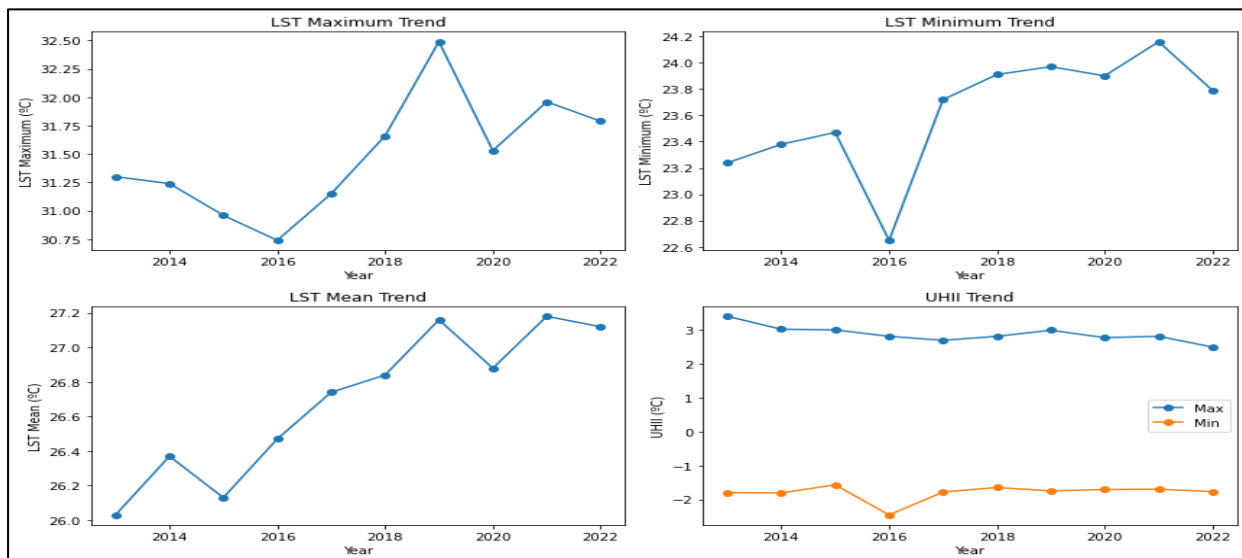


Figure 4.12: Dry Season LST Trends (2013 – 2022)

Table 4.1: Rainy Seasonal Variation in LST

Season	Year	LST Max (°C)	LST Min (°C)	LST Mean (°C)	STD	UHI (°C)	
						Max	Min
Rainy Season	2013	31.75	22.13	25.41	1.86	3.39	-1.75
	2014	33.92	20.44	25.50	2.74	3.36	-1.95
	2015	32.48	20.23	25.22	1.98	3.80	-2.58
	2016	33.57	21.89	26.55	2.35	3.15	-2.06
	2017	33.86	9.29	25.04	2.44	3.67	-6.58
	2018	32.21	16.03	25.26	2.03	3.61	-4.74
	2019	33.11	21.90	25.91	2.38	3.17	-1.75
	2020	33.07	22.10	26.48	2.41	2.91	-1.89
	2021	33.24	20.93	26.47	2.24	3.18	-2.55
	2022	30.89	18.22	24.76	2.01	3.05	-3.38

Table 4.2: Dry Seasonal Variation in LST

Season	Year	LST Max (°C)	LST Min (°C)	LST Mean (°C)	STD	UHI (°C)	
						Max	Min
Dry Season	2013	31.30	23.24	26.03	1.54	3.41	-1.80
	2014	31.24	23.38	26.37	1.73	3.03	-1.81
	2015	30.96	23.47	26.13	1.68	3.01	-1.57
	2016	30.74	22.65	26.47	1.60	2.82	-2.46
	2017	31.15	23.72	26.74	1.73	2.70	-1.78
	2018	31.66	23.91	26.84	1.80	2.82	-1.65
	2019	32.49	23.97	27.16	1.89	3.00	-1.75
	2020	31.53	23.90	26.88	1.78	2.78	-1.71
	2021	31.96	24.16	27.18	1.81	2.82	-1.70
	2022	31.79	23.79	27.12	1.92	2.50	-1.77

4.1.3 Rainy Season NO2 Tropospheric Column Density

NO₂ concentrations are measured in units of 10^{15} molecules/cm². After analysing the data collected during the rainy season, it has been found that the levels of NO₂ (nitrogen dioxide) vary from year to year. The highest values of NO₂ range from 1.56 to 2.02 (10^{15} molecules/cm²), while the lowest values range from 0.47 to 0.63 (10^{15} molecules/cm²). The average values of NO₂ also fluctuate, ranging from 0.94 to 1.12 (10^{15} molecules/cm²). These fluctuations suggest that the concentrations of NO₂ change during the rainy season. Between 2013 and 2019, the mean values of NO₂ remained relatively stable. However, there was a slight decrease in the concentration of NO₂ in 2020, followed by an increase in 2021 and 2022. It is worth mentioning that no significant outliers were observed in the data, and the points follow a consistent pattern without any extreme values.

4.1.4 Dry Season NO2 Tropospheric Column Density

This data presents the levels of NO₂ detected during the Dry Season, measured in units of 10^{15} molecules/cm² for each year. It's imperative to note that NO₂ concentrations vary yearly, with the NO₂ Max values ranging from 2.32 to 2.69 (10^{15} molecules/cm²), the NO₂ Min values ranging from 0.72 to 1.04 (10^{15} molecules/cm²), and the NO₂ Mean values ranging from 1.41 to 1.80 (10^{15} molecules/cm²). These fluctuations suggest that NO₂ levels can be affected by different factors. The mean NO₂ values slightly increased from 2014 to 2016, followed by relatively stable values from 2017 to 2020. In 2021, there will be a slight decrease in the NO₂ Mean, followed by a minor increase in 2022. Several factors can influence the variations in NO₂ concentrations, such as weather conditions, traffic and industrial emissions, and regional air quality regulations.

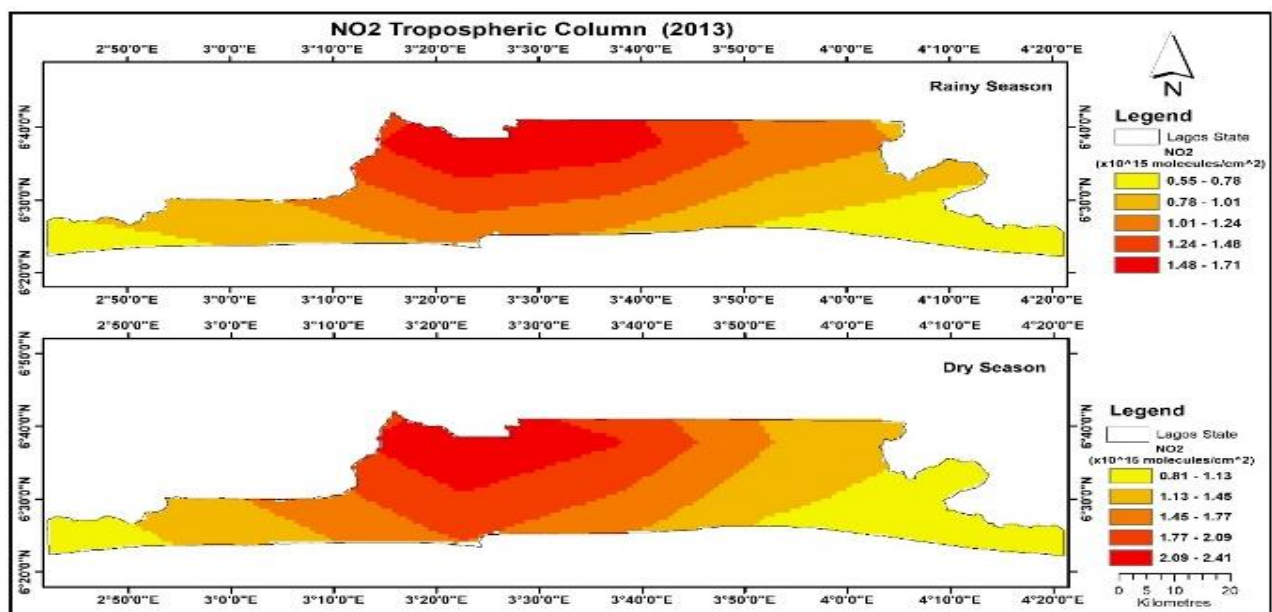


Fig 4.13: NO₂ Tropospheric Column (2013)

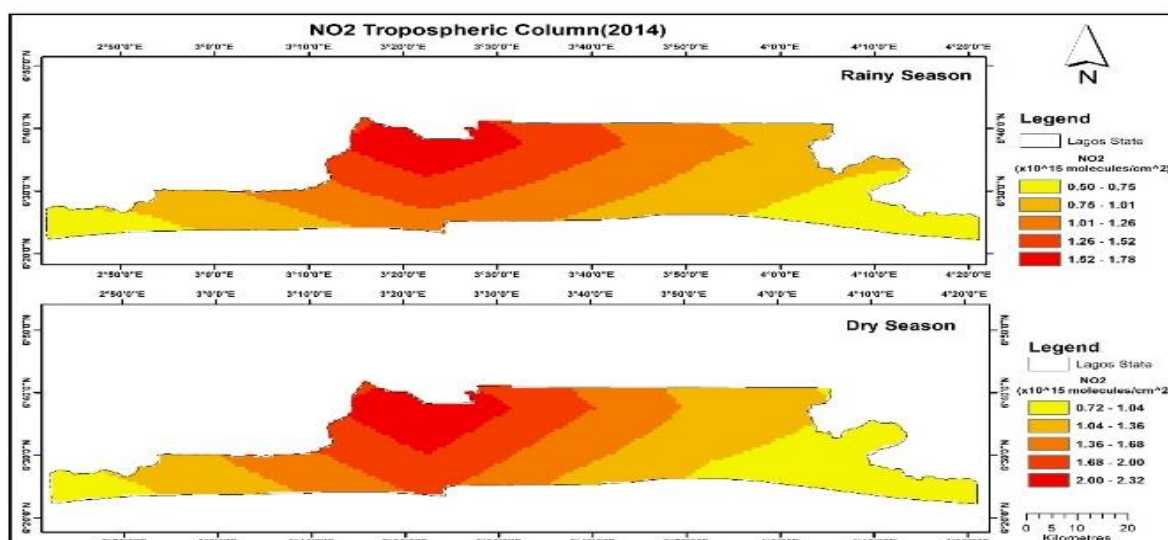


Fig 4.14: NO2 Tropospheric Column (2014)

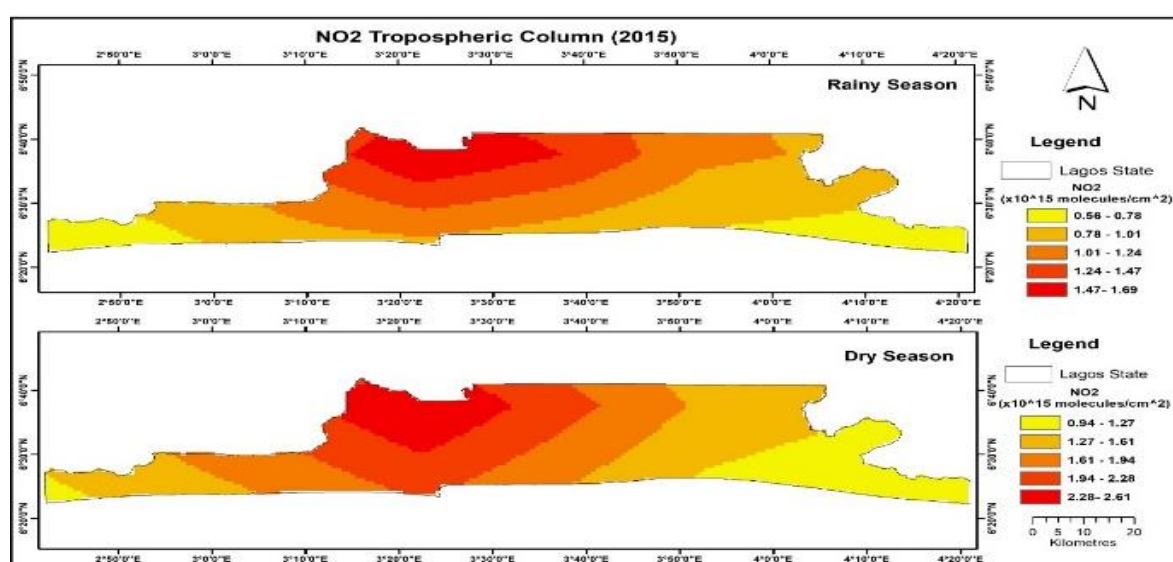


Fig 4.15: NO2 Tropospheric Column (2015)

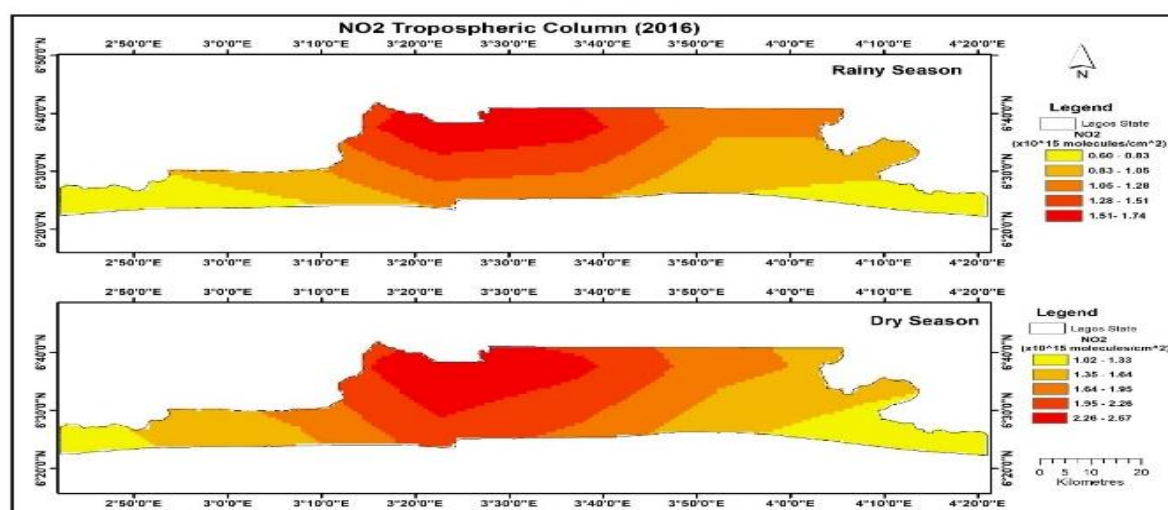


Fig 4.16: NO2 Tropospheric Column (2016)

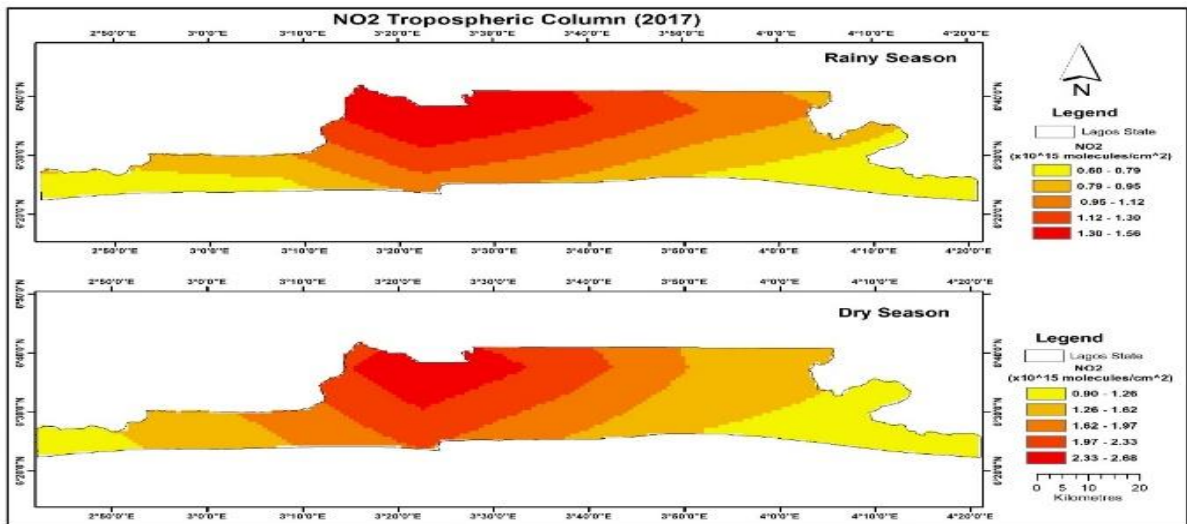


Fig 4.17: NO2 Tropospheric Column (2017)

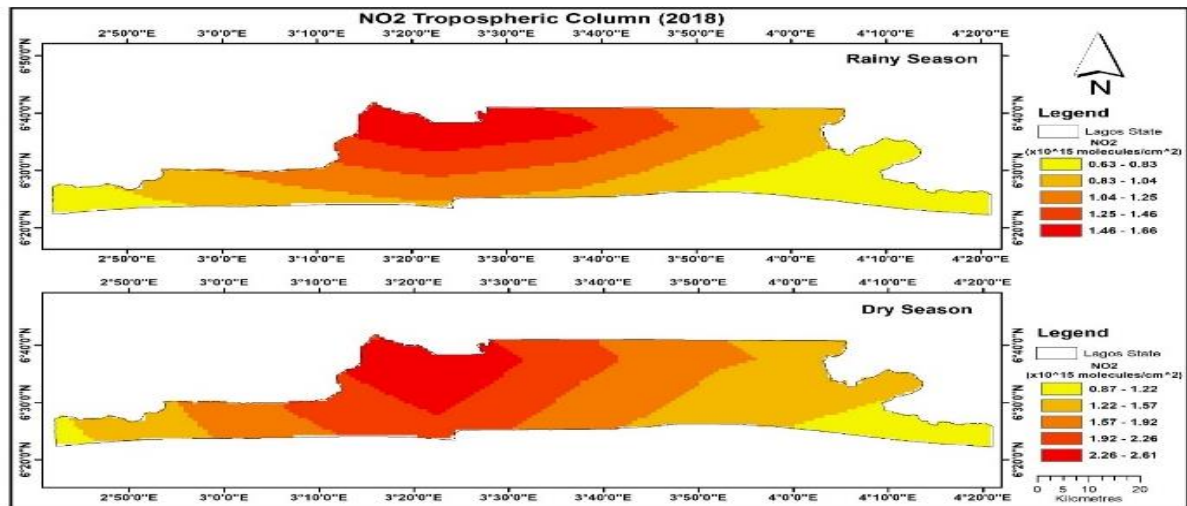


Fig 4.18: NO2 Tropospheric Column (2018)

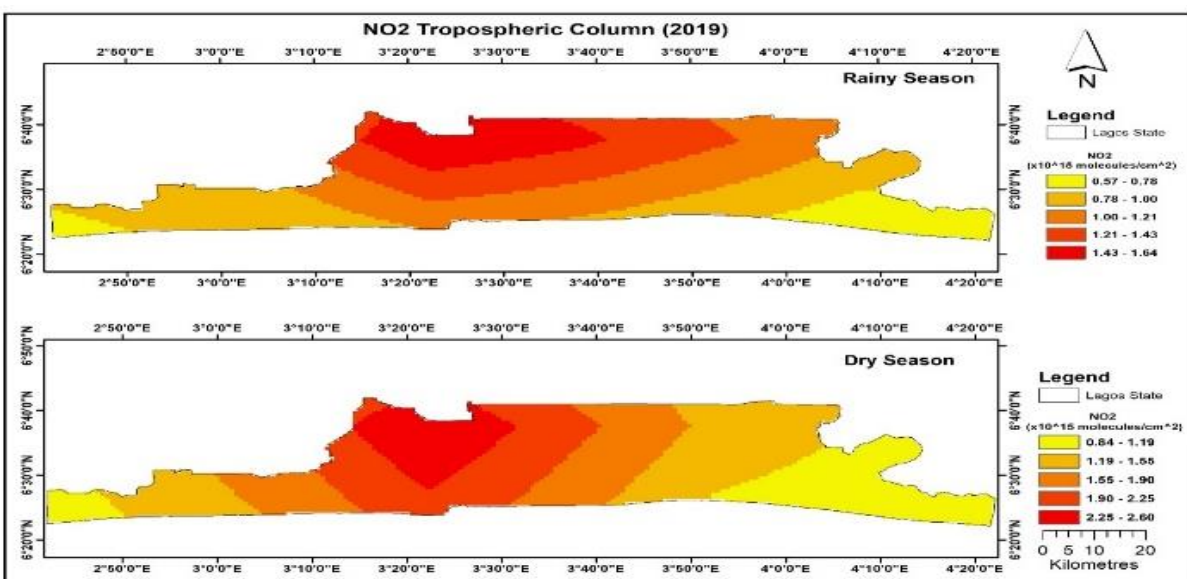


Fig 4.19: NO2 Tropospheric Column (2019)

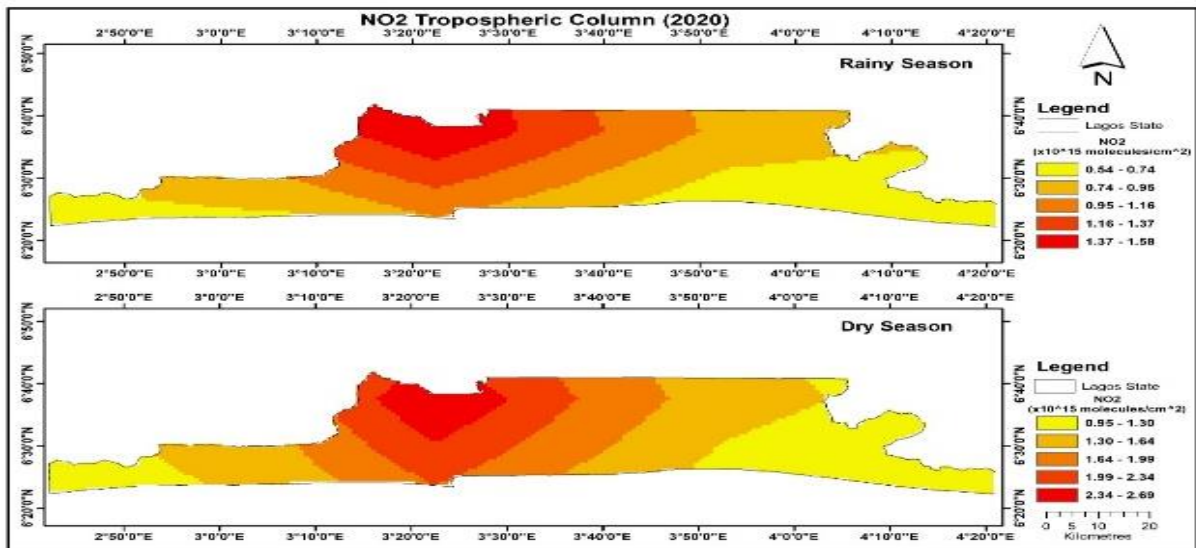


Fig 4.20: NO2 Tropospheric Column (2020)

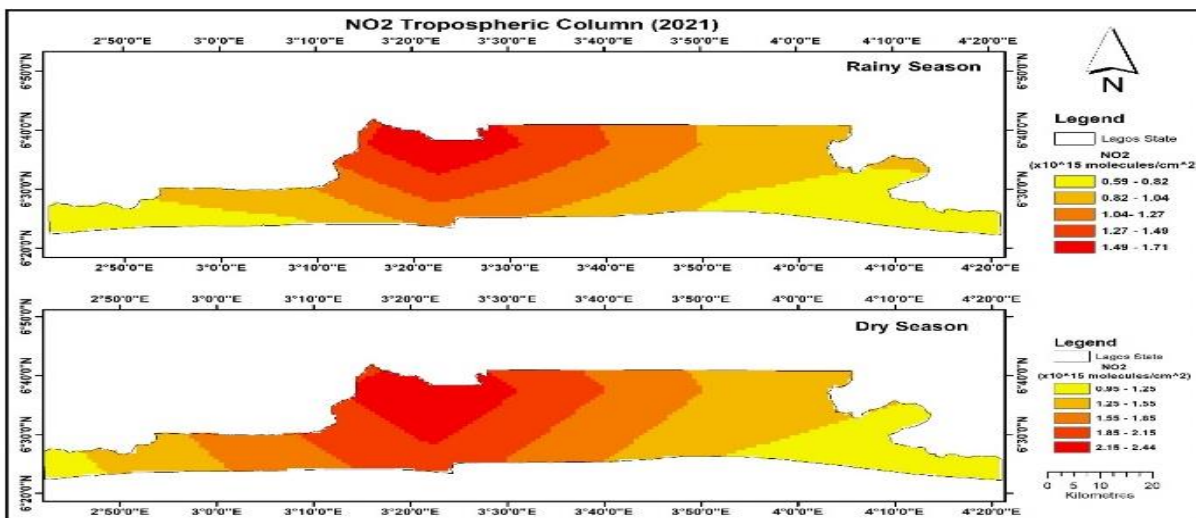


Fig 4.21: NO2 Tropospheric Column (2021)

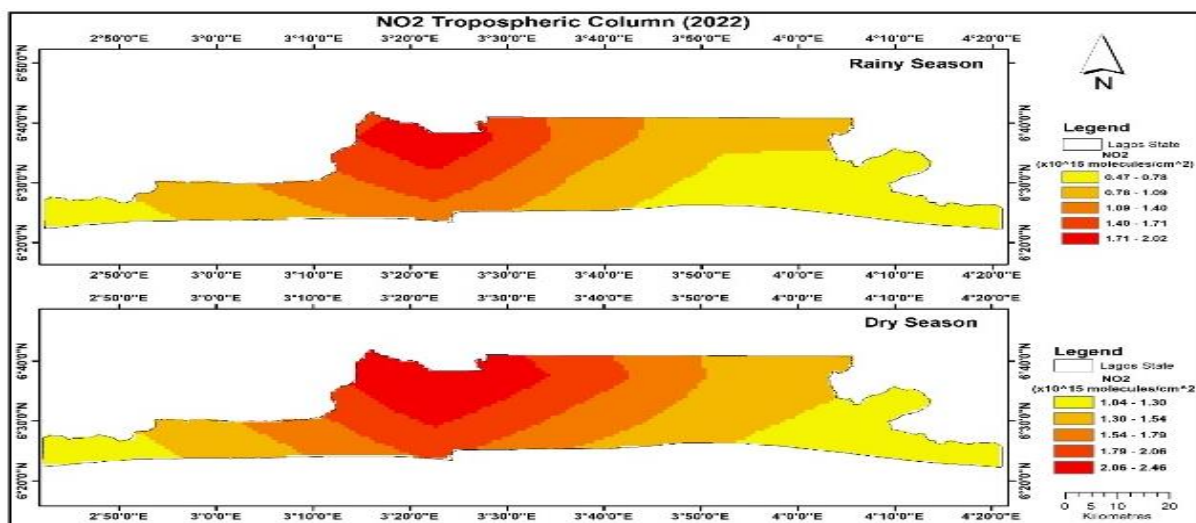


Fig 4.22: NO2 Tropospheric Column (2022)

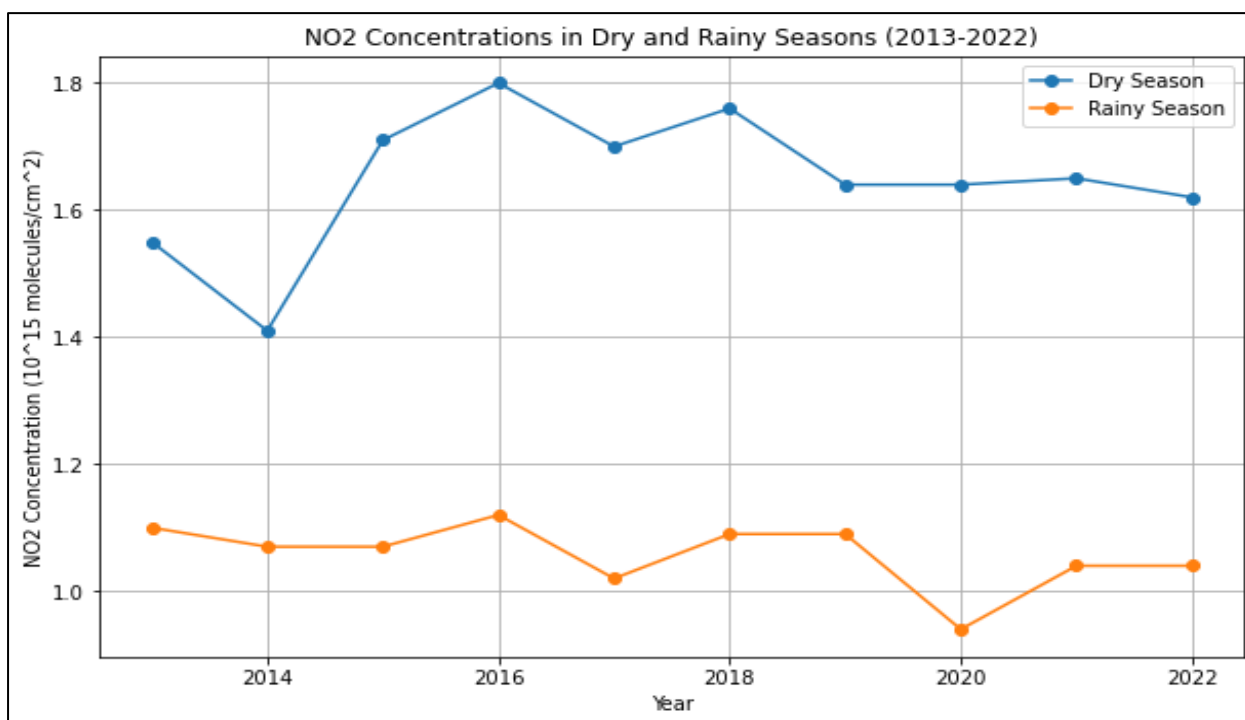


Fig 4.23: NO2 Tropospheric Column Concentrations Trend Profile.

Table 4.3: Seasonal Variation in NO2 Tropospheric Column Density 2013 – 2022

Season	Year	NO2 Max (10 ¹⁵ molecules/cm ²)	NO2 Min (10 ¹⁵ molecules/cm ²)	NO2 Mean (10 ¹⁵ molecules/cm ²)
Dry Season	2013	2.41	0.81	1.55
	2014	2.32	0.72	1.41
	2015	2.61	0.94	1.71
	2016	2.57	1.02	1.80
	2017	2.68	0.90	1.70
	2018	2.61	0.87	1.76
	2019	2.60	0.84	1.64
	2020	2.69	0.95	1.64
	2021	2.44	0.95	1.65
	2022	2.46	1.04	1.62

Table 4.4: Seasonal Variation in NO2 Tropospheric Column Density 2013 – 2022

Season	Year	NO2 Max (10 ¹⁵ molecules/cm ²)	NO2 Min (10 ¹⁵ molecules/cm ²)	NO2 Mean (10 ¹⁵ molecules/cm ²)
Rainy Season	2013	1.71	0.55	1.10
	2014	1.78	0.50	1.07
	2015	1.69	0.56	1.07
	2016	1.74	0.60	1.12
	2017	1.56	0.60	1.02
	2018	1.66	0.63	1.09
	2019	1.64	0.57	1.09
	2020	1.58	0.54	0.94
	2021	1.71	0.59	1.04
	2022	2.02	0.47	1.04

4.2 Correlation between LST and NO2 Tropospheric Column

The correlation coefficient was examined to measure the association between the LST and the concentration of nitrogen dioxide in the troposphere. During the rainy season, the coefficients range from approximately 0.315 to 0.734, while during the dry season, they range from 0.455 to 0.729. Positive values indicate a positive linear relationship between LST values for both seasons. All correlations possess p-values less than 0.001 (See Appendix A), indicating statistical significance and ruling out random chance. The R-squared values are essential for determining the relevance of the linear regression model and eliminating the possibility of chance. These values range from 0.099 to 0.539 for different years and seasons, suggesting that the independent variable can account for 9.9% to 53.9% of the dependent variable's (LST) variance.

The correlation coefficients were generally more substantial during the dry season than the rainy season for each year. Furthermore, During the dry season, the R-squared values were higher compared to the rainy season., implying linear relationship between LST values and the NO2 Tropospheric column density is more robust during the dry season. Overall, the findings signify a positive and statistically significant linear relationship between LST values for each year's rainy and dry seasons. Nonetheless It seems that the association between them is more robust during the dry season compared to the rainy season.. Additionally, the R-squared values suggest that the season (rainy vs. dry) can explain a significant proportion of the variance in LST values for each year.

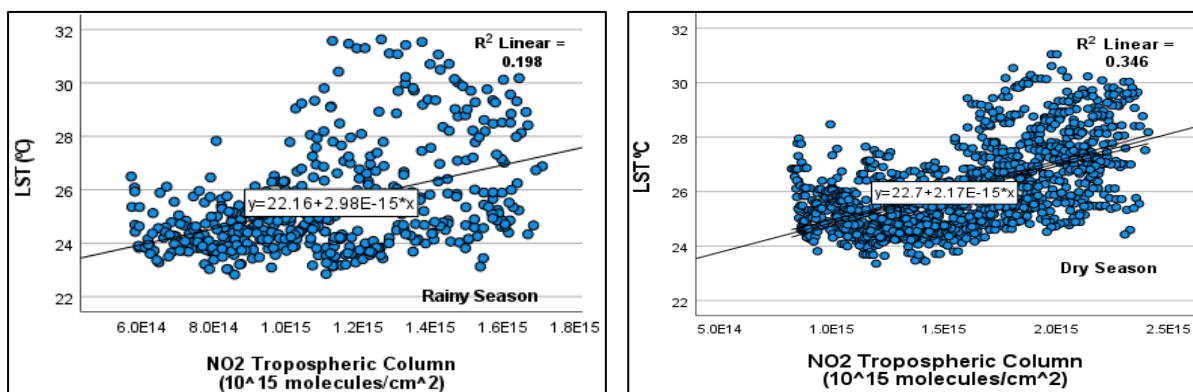


Figure 4.24: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2013.

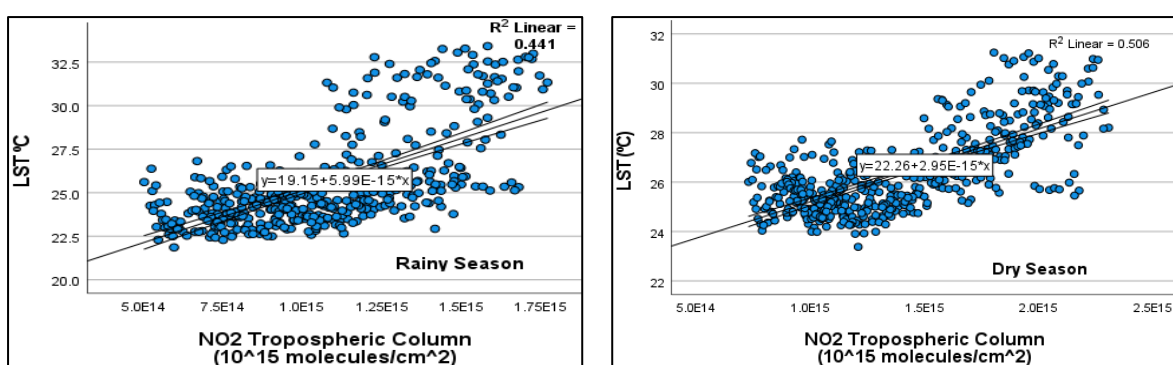


Figure 4.25: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2014.

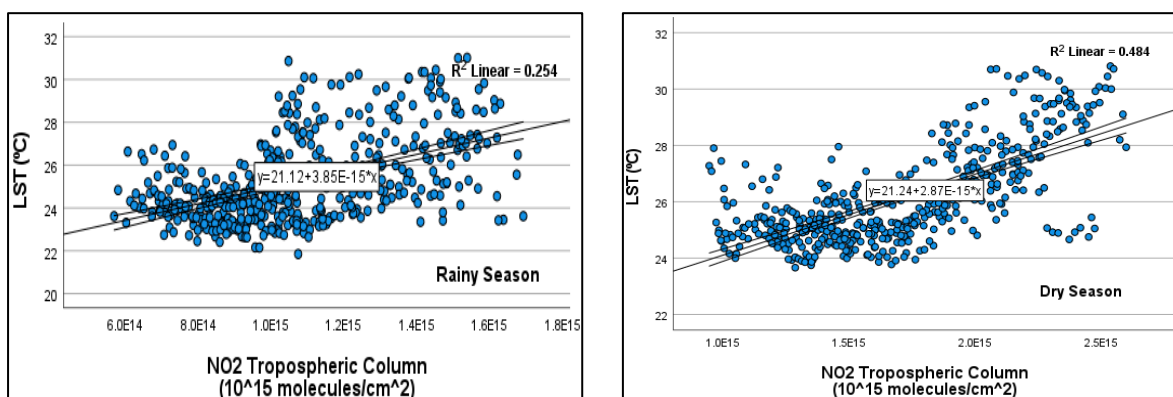


Figure 4.26: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2015

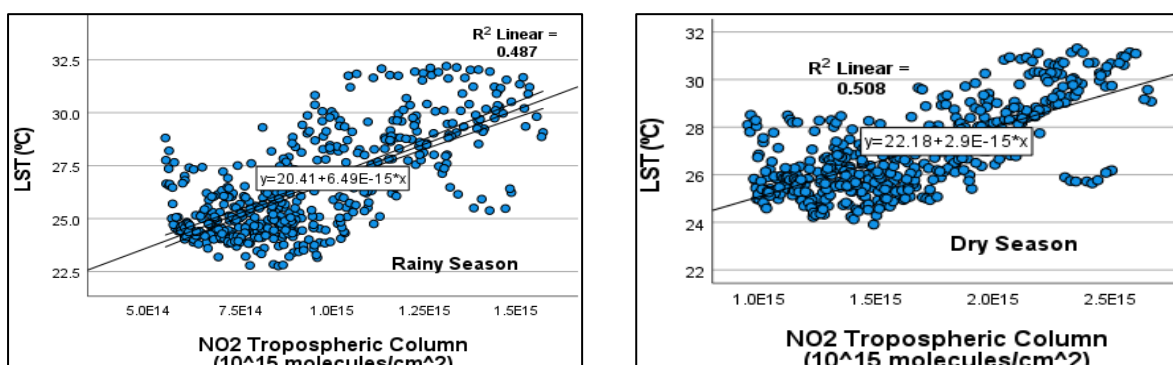


Figure 4.27: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2016.

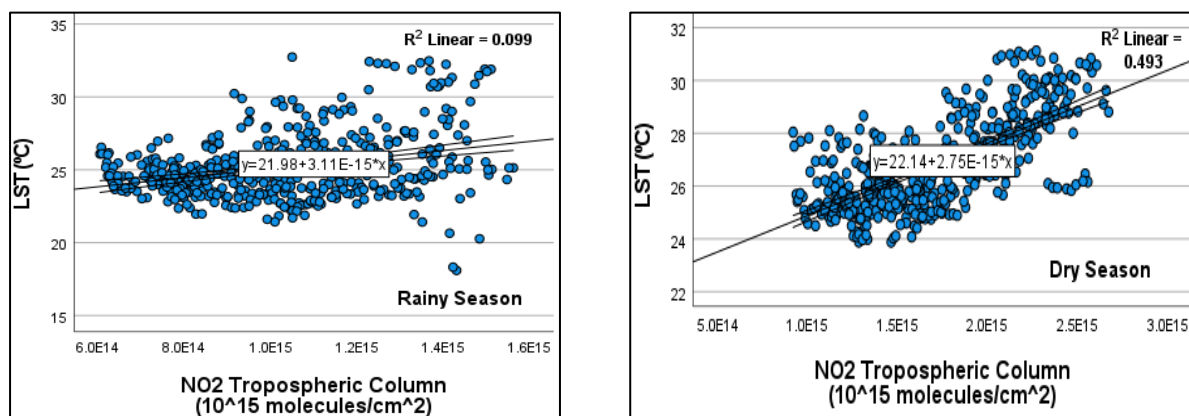


Figure 4.28: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2017.

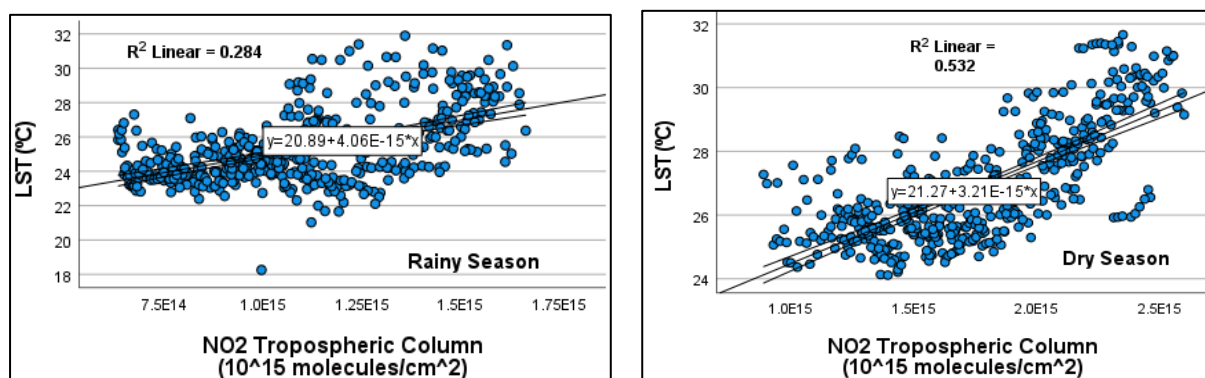


Figure 4.29: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2018.

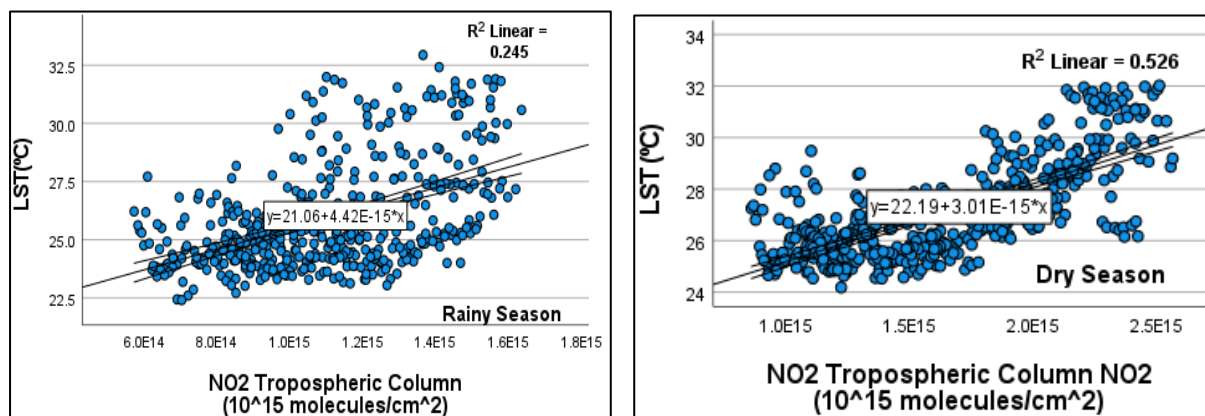


Figure 4.30: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2019.

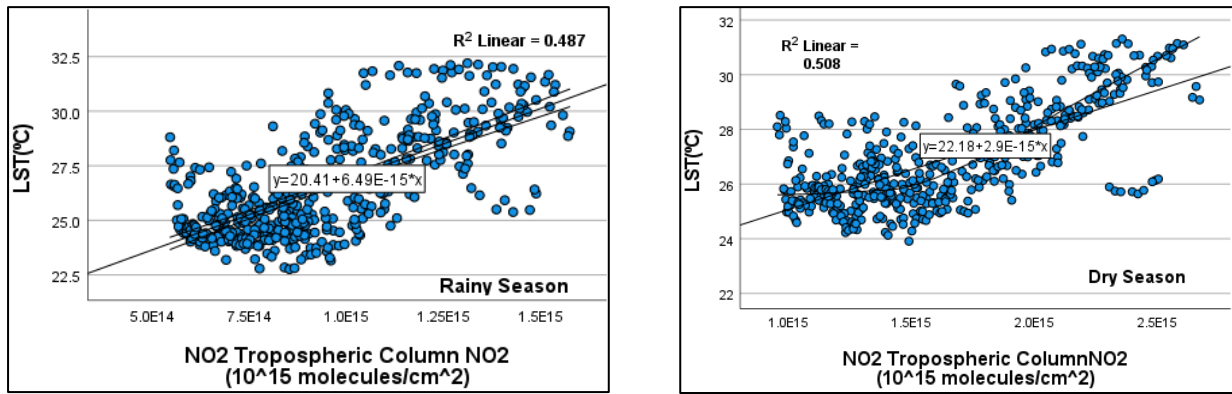


Figure 4.31: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2020.

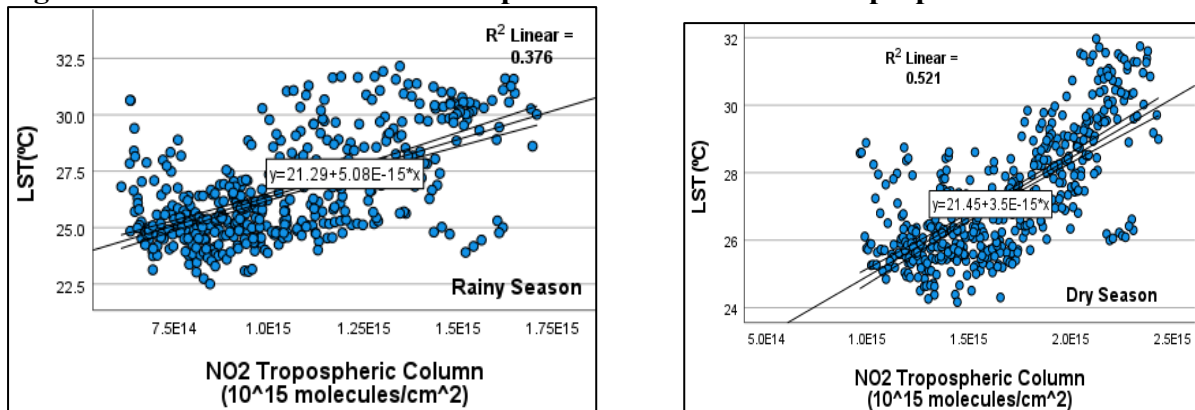


Figure 4.32: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2021.

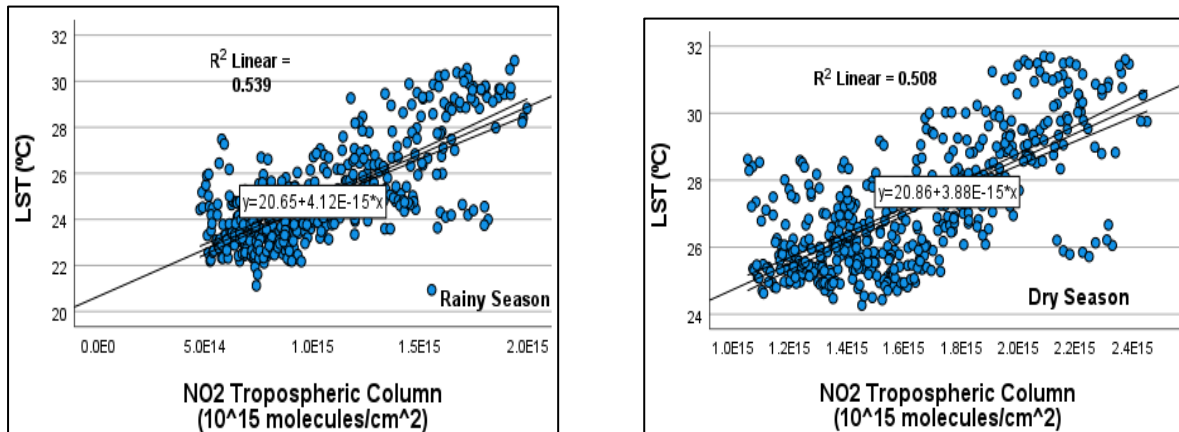


Figure 4.33: Scatter Plot Relationship between LST & NO2 Tropospheric Column 2022.

4.3 Land Use Land Cover Classification (LULCC) 2013 – 2022

The land-use alterations from 2013 to 2022 were examined using a machine-learning algorithm called the support vector machine. Recent research has indicated that SVM is especially powerful when dealing with complex and multi-dimensional datasets. It has been discovered to be more precise than the maximum likelihood classifier (C. Huang, Davis, and Townshend 2002). The outcomes of the LULC classification were authenticated employing a confusion matrix,

which demonstrated an overall accuracy of 82% for 2013 and 87% for 2022. The kappa coefficients were 0.75 and 0.82, respectively (See Tables 4.5 and 4.6).

One intriguing discovery was the change in land use over time. For instance, the developed regions increased by 12.63%, while the vegetative coverage also increased by 5.93%. However, there was a reduction in water bodies and wetlands by approximately 1.23% and 24.26%, respectively. These changes could contribute to the general warming of the city, primarily due to anthropogenic activities.

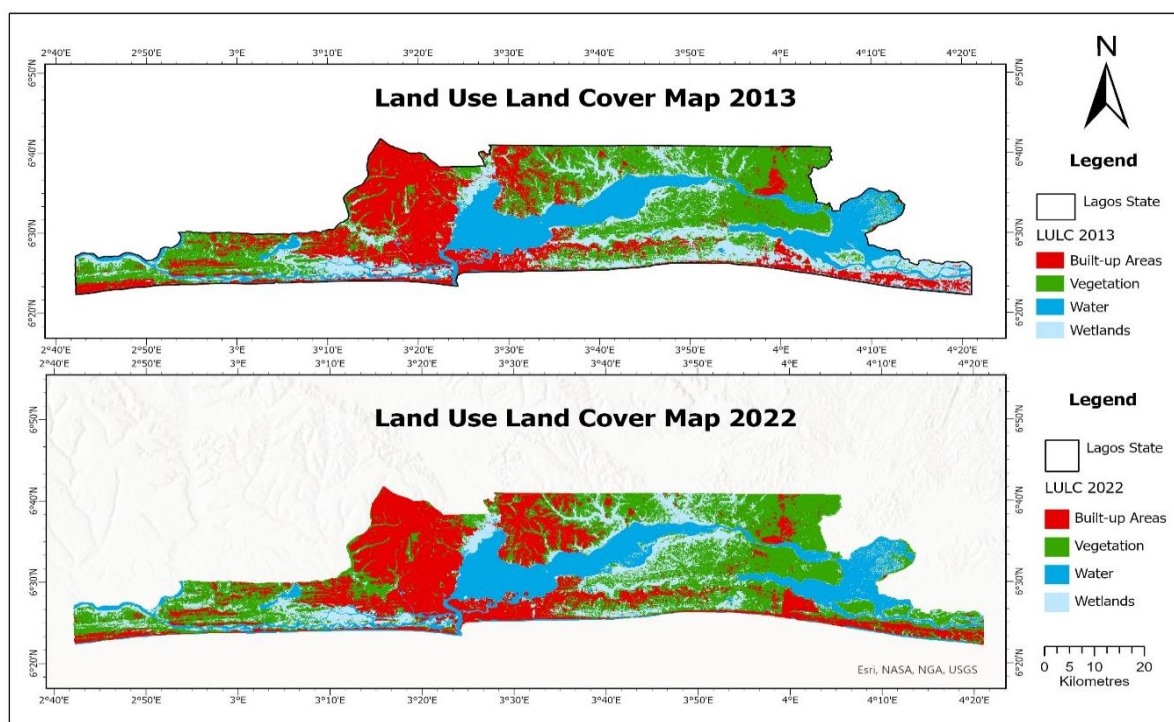


Fig 4.34: Land Use Land Cover (2013 -2022)

Table 4.5: Confusion Matrix for 2013 Satellite Image

Class	Water	Built-up Areas	Vegetation	Wetlands	Total	U-Accuracy	Kappa
Water	20	0	0	0	20	1	0
Built-up Areas	0	25	1	0	26	0.961538462	0
Vegetation	0	7	26	0	33	0.787878788	0
Wetlands	0	2	8	11	21	0.523809524	0
Total	20	34	35	11	100	0	0
P_Accuracy	1	0.735294118	0.7428571	1	0	0.82	0
Kappa	0	0	0	0	0	0	0.754434

Table 4.6: Confusion Matrix for 2022 Satellite Image

Class	Water	Built-up Areas	Vegetation	Wetlands	Total	U-Accuracy	Kappa
Water	18	0	0	2	20	0.9	0
Built-up Areas	0	29	0	0	29	1	0
Vegetation	0	3	26	6	35	0.742857143	0
Wetlands	1	1	0	14	16	0.875	0
Total	19	33	26	22	100	0	0
P_Accuracy	0.94737	0.878787879	1	0.636363636	0	0.87	0
Kappa	0	0	0	0	0	0	0.82434806

Table 4.7: Land Use Land Cover Change (2013 – 2022)

Class	2013 LULC (Sq. Km)	2022 LULC (Sq. Km)	Percentage Change in Land Use
Built-up Areas	942	1061	12.63% (119 Sq. Km)
Vegetation	1214	1286	5.93% (72 Sq. Km)
Water	732	723	1.23% (-9 Sq. Km)
Wetlands	744	562	24.26% (-182 Sq. Km)
Total	3632	3632	

4.4 Mapping of Population at Risk of UHI

This study attempted to identify the population groups most susceptible to urban heat islands (UHI) in Lagos State. Zonal statistics operations generated a detailed map of the at-risk areas. Lagos State comprises 20 local government administrative (LGA) council areas, with most of the population concentrated in the state's central region (Figures 4.35 – 4.36). The study classified high-risk areas into five categories, ranging from low to critical risk. Out of the 20 LGAs, eight were identified as essential areas of risk, namely Surulere, Agege, Ifako-ijaye, Ikeja, Shomolu, Oshodi-Isolo, Ajeromi-Ifelodun, and Mushin. Two areas were identified as high-risk, four as medium-risk, four as moderate-risk, and two as low-risk. These findings are significant as they provide invaluable resources for policymakers to develop effective approaches to abate the effect of UHIs on vulnerable populations.

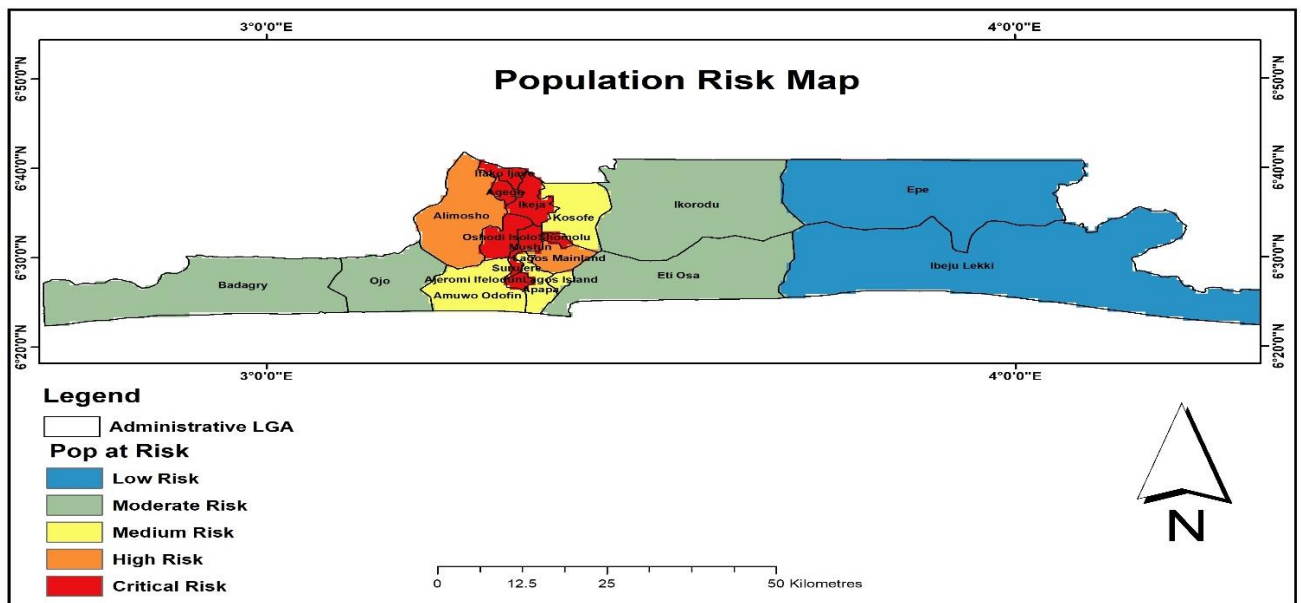


Fig. 4.35: Population at Risk of UHI and the NO₂ Tropospheric Column Effect

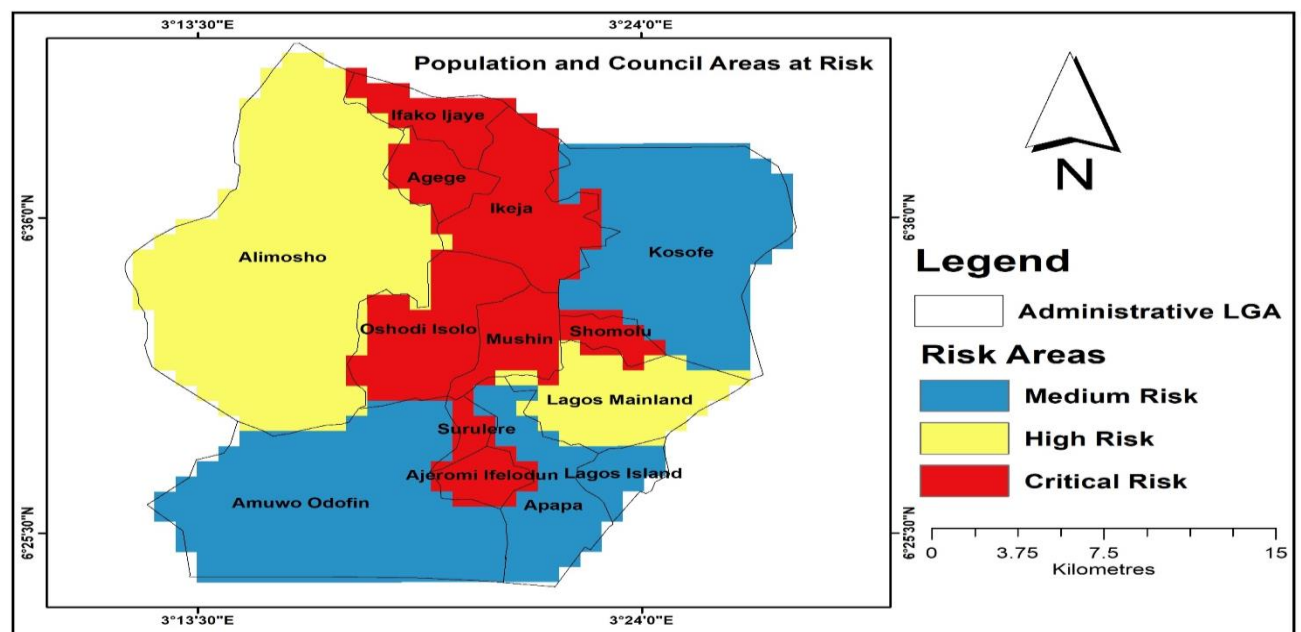


Fig. 4.36: Local Council Areas that Require Intervention

4.5 Limitations of Study.

This study has certain limitations. The MODIS data had some date gaps due to cloud cover and its coarse resolution, but it offers the advantage of year-round coverage, which is a major limitation of Landsat because it is not available throughout all seasons. Additionally, Lagos only has one weather station located at the airport, which does not provide spatial coverage for the entire state. Furthermore, there is a lack of up-to-date population data, as the last known population census in Nigeria was conducted in 2006. Therefore, gridded population estimates was used, which may not provide adequate population figures.

Chapter 5: Discussion, Conclusion and Recommendations

5.1 Discussion

Nigeria can be classified into three different climatic zones: a tropical monsoon climate in the southern part, a tropical savannah climate in most central regions, and a Sahelian hot and semi-arid climate in the northern part. The highest temperatures are usually experienced during the dry season and do not fluctuate significantly from the coastal areas to the inland regions. (Climate Change Knowledge Portal, 2021).

Lagos, the study location, is in the tropical monsoon climatic zone. The rainy season is characterised by significant fluctuations in both temperature and precipitation. Throughout this season, the LST Max, Min, and Mean temperatures exhibit annual variations, reflecting the intensity of rainfall and cloud cover. The LST Mean denotes the average temperature for the duration of the rainy season. It is heavily influenced by both the LST Max and LST Min values. In 2017, there was a striking decline in LST Min during the rainy season, excessive rainfall and cloud cover likely caused the cooler temperatures. The high standard deviation observed in 2017 indicates temperature variability, likely attributed to varying cloud cover and rainfall patterns that affect the temperature range. The dry season is typically characterised by a decrease in precipitation, increased cloud cover, and an increase in temperature compared to the rainy season. However, it is essential to note that while some general trends can be inferred from the data, there is also a certain degree of variability in the climate. This variability is primarily due to yearly fluctuations in Land Surface Temperature, mainly attributed to monsoon intensity variations that affect cloud cover, precipitation, and temperature.

The LST data for both seasons aligns with the expected temperature trends in this type of climate, with the dry season being warmer on average. Monsoon winds, cloud cover, and precipitation drive temperature fluctuations throughout the year. Consequently, the variability in LST highlights dynamic nature of these climates. From 2013 to 2022, the average LST temperature change was 0.76 °C. This change aligns with the average temperature increase observed from 1991 to 2020, indicating that Nigerian states experienced temperatures of 0.5 °C to 1.5 °C higher than the usual long-term mean(Nigerian Meteorological Agency, 2021).

It is crucial to research further and monitor these changes better to understand their impact on the environment and human health. The monsoon climate can also significantly impact the concentration of nitrogen dioxide (NO₂), which has environmental and public health

implications. During the dry season, weaker monsoon winds can lead to reduced dispersion of pollutants, including NO₂, resulting in increased concentrations near the surface. Stagnant air conditions can also prevent the diffusion of impurities, contributing to higher localised concentrations of NO₂.

However, the rainy season can positively impact air quality by acting as a natural air purifier. Rainfall helps to wash pollutants out of the atmosphere, including NO₂, leading to a temporary decrease in NO₂ concentrations. Heavy rain and increased atmospheric moisture can dilute the concentration of contaminants in the air, further reducing NO₂ levels during and immediately after rain events. Rainwater can also react with NO₂ to form nitric acid, removing it from the atmosphere through wet deposition and further reducing NO₂ concentrations.

The monsoon climate can also impact NO₂ concentration, particularly throughout the dry and rainy seasons, affecting the environment and population health. During the dry season, weaker monsoon winds can lead to reduced dispersion of pollutants, including NO₂, resulting in increased concentrations near the surface. Stagnant air conditions can prevent the diffusion of impurities, contributing to higher localised concentrations of NO₂. Rainy season can positively impact air quality by acting as a natural purifier. Rainfall helps to wash pollutants out of the atmosphere, including NO₂, leading to a temporary decrease in NO₂ concentrations. Heavy rain and increased atmospheric moisture can dilute the concentration of contaminants in the air, further reducing NO₂ levels during and immediately after rain events. Rainwater can react with NO₂ to form nitric acid, removing it from the atmosphere through wet deposition and further reducing NO₂ concentrations.

Through the implementation of zonal statistics operations, an in-depth map was created to identify at-risk areas within Lagos State. This map covered 20 local government administrative (LGA) council areas, providing a comprehensive view of the entire region. Figures 4.35-4.36 were utilised to delineate the population at risk within the central region of Lagos State, providing critical information for further UHI assessment. The risk areas identified within the study were categorised into groups, with risk levels ranging from low to critical. In particular, it was revealed that 8 LGAs were in dire need of urgent evaluation regarding UHI risk in the population.

This study holds great significance for policymakers. Identifying vulnerable areas and population groups enables policymakers to effectively allocate resources and develop targeted strategies to mitigate the impact of UHIs. By focusing on critical areas, authorities can

implement measures to provide relief, improve urban planning, enhance green spaces, and implement heat reduction strategies. These measures are designed to protect the most vulnerable populations from the adverse effects of UHI. The findings of this study offer valuable insights into Lagos State's urban planning and climate adaptation efforts. The categorised risk levels provide a framework for prioritising interventions and resource allocation, ensuring that areas with the highest vulnerability to UHI receive the necessary attention. With this framework in place, policymakers can take proactive measures to address the impact of UHIs and work towards creating a safer, more sustainable future for the people of Lagos State.

5.2 Conclusion

Nigeria's climate characteristics, specifically Lagos State, and the significance of the study in identifying areas vulnerable to urban heat islands (UHIs). Lagos, is located in the tropical monsoon climatic zone, experiences distinct rainy and dry seasons, resulting in fluctuations in temperature and precipitation.

The study's focus on Lagos State provides valuable insights into the temperature variations during rainy and dry seasons. The data aligns with expected temperature trends for this climate type, with the dry season being warmer on average. The variability in Land Surface Temperature values highlights the dynamic nature of these climates. The calculated average LST temperature change from 2013 to 2022 is approximately 0.76 °C, corresponding to the observed average temperature (0.5 °C – 1.5 °C) increase from 1991 to 2020. This information underscores the importance of continued research and monitoring to understand these changes.

Applying zonal statistics operations to create a detailed map of at-risk areas within Lagos State is essential for identifying vulnerable areas, such as the eight local government administrative (LGA) council areas classified as a critical risk. This empowers policymakers to allocate resources effectively and intervene on the most vulnerable populations. By prioritising interventions based on categorised risk levels, authorities can focus on areas that require immediate attention for heat reduction strategies, providing valuable insights for adaptation efforts in Lagos State and highlighting climate dynamics, especially the influences of UHIs on vulnerable people.

5.3 Recommendation

In Lagos State, Nigeria, only one weather station is available, and it is situated at the Airport. To conduct more thorough research on Land Surface Temperature, it is essential to carry out multi-sensor LST studies. This is due to the limited spatial coverage resulting from the absence of weather stations within the state. It is also necessary to closely monitor changes to continually assess and evaluate the impact of LST and air pollutants on human health. Areas that have been identified require further evaluation to assess the vulnerable population that may require urgent intervention. Finally, it is important to prioritise urban greening as a matter of urgency.

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Appendices

Appendix A: Python Code for Retrieving and Processing MODIS Aqua MYD11A2 Data.

```
# -*- coding: utf-8 -*-
"""
@author: Oluwaseun
"""

#import libraries
import arcpy
import os
from arcpy.sa import *
from arcpy.ia import *

#Enlist the HDF files from the raw data folder
arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/'
hdfList = arcpy.ListRasters()
print(hdfList)

#Extract the daytime temperature layer
os.mkdir('/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Day/')
rootPath = '/MODIS_AquaSeasonal/2013/RainySeason2013/'
outputPath = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Day/'
for filename in hdfList:
    print(rootPath+filename)
    arcpy.ExtractSubDataset_management(in_raster=rootPath+filename, out_raster=outputPath+filename)

#Extract the Night-time temperature layer
os.mkdir('/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Night/')
rootPath = '/MODIS_AquaSeasonal/2013/RainySeason2013/'
outputPath = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Night/'
for filename in hdfList:
    print(rootPath+filename)
    arcpy.ExtractSubDataset_management(in_raster=rootPath+filename, out_raster=outputPath+filename)

#Create mean of daytime and night temperature layer
arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Day/'
Aqua_Day = arcpy.ListRasters("*.TIF")
print(Aqua_Day)

#os.mkdir('/MODIS_AquaSeasonal/2013/DrySeason2013/AquaMean_Day/')
arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Night/'
Aqua_Night = arcpy.ListRasters("*.TIF")
print(Aqua_Night)
os.mkdir('/MODIS_AquaSeasonal/2013/RainySeason2013/AquaMean/')
for (i,j) in zip(Aqua_Day, Aqua_Night):
    outCellStats = CellStatistics([f'/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Day/{i}',
    f'/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Night/{j}'],
    f'/MODIS_AquaSeasonal/2013/RainySeason2013/AquaMean/{i}')

#outCellStats.save(f'/MODIS_AquaSeasonal/2013/DrySeason2013/AquaMean_Night/{j}')
arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/AquaMean/'
rootPath = '/MODIS_AquaSeasonal/2013/RainySeason2013/AquaMean/'
os.mkdir('/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/')
outputPath = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/'
rasterList = arcpy.ListRasters("*.TIF")
for filename in master list:
```

```

print (rootPath+filename)
output_raster = arcpy.sa.Raster(filename) * 0.02 - 273.15
output_raster.save(outputPath+filename)

```

Calculating Cell Statistics for MEAN.

```

arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/'
Cel_rasterList = arcpy.ListRasters("*", "TIF")
outCellStats = CellStatistics(Cel_rasterList, "MEAN", "DATA")
outCellStats.save("/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/LST.tif")

```

Calculating Cell Statistics for MAXIMUM.

```

arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/'
Cel_rasterList = arcpy.ListRasters("*", "TIF")
outCellStats = CellStatistics(Cel_rasterList, "MAXIMUM", "DATA")
outCellStats.save("/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/LSTmax.tif")

```

Calculating Cell Statistics for MINIMUM.

```

arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/'
Cel_rasterList = arcpy.ListRasters("*", "TIF")
outCellStats = CellStatistics(Cel_rasterList, "MINIMUM", "DATA")
outCellStats.save("/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/LSTmin.tif")

```

Calculating Cell Statistics for STD.

```

arcpy.env.workspace = '/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/'
Cel_rasterList = arcpy.ListRasters("*", "TIF")
outCellStats = CellStatistics(Cel_rasterList, "STD", "DATA")
outCellStats.save("/MODIS_AquaSeasonal/2013/RainySeason2013/Aqua_Celsius/LSTstd.tif")

```

Appendix B: Python Code for Retrieving and Processing NO2 Tropospheric Column Data

```
# -*- coding: utf-8 -*-
"""
@author: Oluwaseun
"""

#import libraries
import arcpy
import os
from arcpy.sa import *
from arcpy.ia import *

#Enlist the He5 files from the raw data folder
arcpy.env.workspace = '/OMI_AURA/NO2_2022/NO2_DrySeason/'
he5List = arcpy.ListRasters()
print(he5List)

#Extract NO2 Total Column 30% Cloud Screened
os.mkdir ('/OMI_AURA/NO2_2022/NO2_DrySeason/NO2/')
rootPath = '/OMI_AURA/NO2_2022/NO2_DrySeason/'
outputPath = '/OMI_AURA/NO2_2022/NO2_DrySeason/NO2/'
for filename in he5List:
    print(rootPath+filename)
    arcpy.ExtractSubDataset_management(in_raster=rootPath+filename, out_raster=outputPath+filename[

#Extract the NO2 Tropospheric Column
os.mkdir ('/OMI_AURA/NO2_2022/NO2_DrySeason/NO2Trop_2022/')
rootPath = '/OMI_AURA/NO2_2022/NO2_DrySeason/'
outputPath = '/OMI_AURA/NO2_2022/NO2_DrySeason/NO2Trop_2022/'
for filename in he5List:
    print(rootPath+filename)
    arcpy.ExtractSubDataset_management(in_raster=rootPath+filename, out_raster=outputPath+filename[

# Calculating Cell Statistics Mean NO2 Total Column
arcpy.env.workspace = '/OMI_AURA/NO2_2022/NO2_DrySeason/NO2/'
NO2_rasterList = arcpy.ListRasters("*.TIF")
outCellStats = CellStatistics(NO2_rasterList, "MEAN", "DATA")
outCellStats.save("/OMI_AURA/NO2_2022/NO2_DrySeason/NO2/NO2.tif")

# Calculating Cell Statistics Mean NO2 Tropospheric Column
arcpy.env.workspace = '/OMI_AURA/NO2_2022/NO2_DrySeason/NO2Trop_2022/'
NO2Trop_rasterList = arcpy.ListRasters("*.TIF")
outCellStats = CellStatistics(NO2Trop_rasterList, "MEAN", "DATA")
outCellStats.save("/OMI_AURA/NO2_2022/NO2_DrySeason/NO2trop.tif")

#Enlist the HE5 files from the raw data folder for rainy season period
arcpy.env.workspace = '/OMI_AURA/NO2_2022/NO2_RainySeason/'
he5List = arcpy.ListRasters()
print(he5List)

#Extraction NO2 Total Column density
os.mkdir ('/OMI_AURA/NO2_2022/NO2_RainySeason/NO2/')
rootPath = '/OMI_AURA/NO2_2022/NO2_RainySeason/'
outputPath = '/OMI_AURA/NO2_2022/NO2_RainySeason/NO2/'
for filename in he5List:
```

```

print(rootPath+filename)
arcpy.ExtractSubDataset_management(in_raster=rootPath+filename, out_raster=outputPath+filename[

#Extraction NO2 Trophopheric Column density
os.mkdir ('/OMI_AURA/NO2_2022/NO2_RainySeason/NO2r/')
rootPath = '/OMI_AURA/NO2_2022/NO2_RainySeason/'
outputPath = '/OMI_AURA/NO2_2022/NO2_RainySeason/NO2r/'
for filename in he5List:
print(rootPath+filename)
arcpy.ExtractSubDataset_management(in_raster=rootPath+filename, out_raster=outputPath+filename[

# Calculating Cell Statistics Mean NO2 Total Column.
arcpy.env.workspace = '/OMI_AURA/NO2_2022/NO2_RainySeason/NO2/'
NO2_rasterList = arcpy.ListRasters("*", "TIF")
outCellStats = CellStatistics(NO2_rasterList, "MEAN", "DATA")
outCellStats.save("/OMI_AURA/NO2_2022/NO2_RainySeason/NO2/NO2r.tif")

# Calculating Cell Statistics NO2 Trophopheric Column
arcpy.env.workspace = '/OMI_AURA/NO2_2022/NO2_RainySeason/NO2r/'
NO2Trop_rasterList = arcpy.ListRasters("*", "TIF")
outCellStats = CellStatistics(NO2Trop_rasterList, "MEAN", "DATA")
outCellStats.save("/OMI_AURA/NO2_2022/NO2_RainySeason/NO2r/NO2tr.tif")

```

Appendix C: Python Code for LST (Dry and Rainy Season) Trend Profile.

```
# -*- coding: utf-8 -*-
"""
@author: Oluwaseun
"""

import matplotlib.pyplot as plt
years = [2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022]
LST_Maximum = [31.30, 31.24, 30.96, 30.74, 31.15, 31.66, 32.49, 31.53, 31.96, 31.79]
LST_Minimum = [23.24, 23.38, 23.47, 22.65, 23.72, 23.91, 23.97, 23.90, 24.16, 23.79]
LST_Mean = [26.03, 26.37, 26.13, 26.47, 26.74, 26.84, 27.16, 26.88, 27.18, 27.12]
STD = [1.54, 1.73, 1.68, 1.60, 1.73, 1.80, 1.89, 1.78, 1.81, 1.92]
UHII_Max = [3.41, 3.03, 3.01, 2.82, 2.70, 2.82, 3.00, 2.78, 2.82, 2.50]
UHII_Min = [-1.80, -1.81, -1.57, -2.46, -1.78, -1.65, -1.75, -1.71, -1.70, -1.77]
plt.figure(figsize=(12, 8))

# Plot LST Maximum
plt.subplot(2, 2, 1)
plt.plot(years, LST_Maximum, marker='o')
plt.xlabel("Year")
plt.ylabel("LST Maximum (°C)")
plt.title("LST Maximum Trend")

# Plot LST Minimum
plt.subplot(2, 2, 2)
plt.plot(years, LST_Minimum, marker='o')
plt.xlabel("Year")
plt.ylabel("LST Minimum (°C)")
plt.title("LST Minimum Trend")

# Plot LST Mean
plt.subplot(2, 2, 3)
plt.plot(years, LST_Mean, marker='o')
plt.xlabel("Year")
plt.ylabel("LST Mean (°C)")
plt.title("LST Mean Trend")

# Plot of UHII Maximum and Minimum
plt.subplot(2, 2, 4)
plt.plot(years, UHII_Max, marker='o', label='Max')
plt.plot(years, UHII_Min, marker='o', label='Min')
plt.xlabel("Year")
plt.ylabel("UHII (°C)")
plt.title("UHII Trend")
plt.legend()
plt.tight_layout()
plt.show()
import numpy as np
from scipy.stats import linregress
def trend_analysis(data):
    x = np.arange(len(data))
    slope, _, r_value, p_value, _ = linregress(x, data)
```

```

return slope, r_value, p_value
trend_LST_Maximum, r_value_LST_Maximum, p_value_LST_Maximum = trend_analysis(LST_Maximum)
trend_LST_Minimum, r_value_LST_Minimum, p_value_LST_Minimum = trend_analysis(LST_Minimum)
trend_LST_Mean, r_value_LST_Mean, p_value_LST_Mean = trend_analysis(LST_Mean)
trend_STD, r_value_STD, p_value_STD = trend_analysis(STD)
trend_UHII_Max, r_value_UHII_Max, p_value_UHII_Max = trend_analysis(UHII_Max)
trend_UHII_Min, r_value_UHII_Min, p_value_UHII_Min = trend_analysis(UHII_Min)

print("Trends:")
print(f"LST Maximum: {trend_LST_Maximum:.3f} °C per year (p-value: {p_value_LST_Maximum:.3f})")
print(f"LST Minimum: {trend_LST_Minimum:.3f} °C per year (p-value: {p_value_LST_Minimum:.3f})")
print(f"LST Mean: {trend_LST_Mean:.3f} °C per year (p-value: {p_value_LST_Mean:.3f})")
print(f"STD: {trend_STD:.3f} per year (p-value: {p_value_STD:.3f})")
print(f"UHII Maximum: {trend_UHII_Max:.3f} °C per year (p-value: {p_value_UHII_Max:.3f})")
print(f"UHII Minimum: {trend_UHII_Min:.3f} °C per year (p-value: {p_value_UHII_Min:.3f})")

```

Appendix D: Python Code for NO2 Tropospheric Column Trend Profile.

```
# -*- coding: utf-8 -*-

"""
@author: Oluwaseun
"""

import matplotlib.pyplot as plt
# NO2 Data for Dry Season and Rainy Season
years = [2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022]
no2_dry = [1.55, 1.41, 1.71, 1.80, 1.70, 1.76, 1.64, 1.64, 1.65, 1.62]
no2_rainy = [1.10, 1.07, 1.07, 1.12, 1.02, 1.09, 1.09, 0.94, 1.04, 1.04]
# Create a figure and axis
plt.figure(figsize=(10, 6))
plt.plot(years, no2_dry, marker='o', label='Dry Season')
plt.plot(years, no2_rainy, marker='o', label='Rainy Season')
# Plot trends analysis label
plt.title('NO2 Concentrations in Dry and Rainy Seasons (2013-2022)')
plt.xlabel('Year')
plt.ylabel('NO2 Concentration (10^15 molecules/cm^2)')
plt.legend()
plt.grid(True)
# Show the trend analysis plot
plt.show()
```

Appendix E: Pearson Correlation

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.588**
	Sig. (2-tailed)		<.001
	N	1311	1311
NO2 Tropospheric Column	Pearson Correlation	.588**	1
	Sig. (2-tailed)	<.001	
	N	1311	1311

** . Correlation is significant at the 0.01 level (2-tailed).

2013

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.712**
	Sig. (2-tailed)		<.001
	N	475	475
NO2 Tropospheric Column	Pearson Correlation	.712**	1
	Sig. (2-tailed)	<.001	
	N	475	475

** . Correlation is significant at the 0.01 level (2-tailed).

2014

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.504**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.504**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

2015

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.698**
	Sig. (2-tailed)		<.001
	N	478	478
NO2 Tropospheric Column	Pearson Correlation	.698**	1
	Sig. (2-tailed)	<.001	
	N	478	478

** . Correlation is significant at the 0.01 level (2-tailed).

2016

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.445**
	Sig. (2-tailed)		<.001
	N	477	477
NO2 Tropospheric Column	Pearson Correlation	.445**	1
	Sig. (2-tailed)	<.001	
	N	477	477

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.664**
	Sig. (2-tailed)		<.001
	N	475	475
NO2 Tropospheric Column	Pearson Correlation	.664**	1
	Sig. (2-tailed)	<.001	
	N	475	475

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.696**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.696**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.713**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.713**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.315**
	Sig. (2-tailed)		<.001
	N	478	478
NO2 Tropospheric Column	Pearson Correlation	.315**	1
	Sig. (2-tailed)	<.001	
	N	478	478

** . Correlation is significant at the 0.01 level (2-tailed).

2017

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.702**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.702**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.533**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.533**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

2018

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.729**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.729**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.495**
	Sig. (2-tailed)		<.001
	N	471	471
NO2 Tropospheric Column	Pearson Correlation	.495**	1
	Sig. (2-tailed)	<.001	
	N	471	471

** . Correlation is significant at the 0.01 level (2-tailed).

2019

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.725**
	Sig. (2-tailed)		<.001
	N	471	471
NO2 Tropospheric Column	Pearson Correlation	.725**	1
	Sig. (2-tailed)	<.001	
	N	471	471

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.698**
	Sig. (2-tailed)		<.001
	N	478	478
NO2 Tropospheric Column	Pearson Correlation	.698**	1
	Sig. (2-tailed)	<.001	
	N	478	478

** . Correlation is significant at the 0.01 level (2-tailed).

2020

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.713**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.713**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.613**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.613**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

2021

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.722**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.722**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.734**
	Sig. (2-tailed)		<.001
	N	479	479
NO2 Tropospheric Column	Pearson Correlation	.734**	1
	Sig. (2-tailed)	<.001	
	N	479	479

** . Correlation is significant at the 0.01 level (2-tailed).

2022

Correlations			
		LST	NO2 Tropospheric Column
LST	Pearson Correlation	1	.713**
	Sig. (2-tailed)		<.001
	N	476	476
NO2 Tropospheric Column	Pearson Correlation	.713**	1
	Sig. (2-tailed)	<.001	
	N	476	476

** . Correlation is significant at the 0.01 level (2-tailed).

Appendix F: Science Data Sets for MODIS/Aqua Land Surface Temperature/Emissivity 8-Day L3 Global 1km SIN Grid V006 (MYD11A2) Science Data Sets.

SDS (hdf - Layers)	Units	Bit Type	Fill	Valid Range	Scale Factor
LST Day:8-Day daytime 1km grid land surface temperature	Kelvin	16-bit unsigned integer	0	7500-65535	0.02
Quality control for daytime LST and emissivity	Bit Field	8-bit unsigned integer	QA	0 -255	NA
Average time of daytime land surface temperature observation	Hours	8-bit unsigned integer	255	0 - 240	0.1
Average view zenith angle of daytime land surface temperature	Degree	8-bit unsigned integer	255	0 -130	1(-65)
LST Night: 8-Day nighttime 1km grid land surface temperature	Kelvin	8-bit unsigned integer	0	7500-65535	0.02
Quality control for nighttime LST and emissivity	Hours	8-bit unsigned integer	QA	0 -255	NA
Average time of nighttime land surface temperature observation	Degree	8-bit unsigned integer	255	0 - 240	0.1
Average view zenith angle of nighttime land surface temperature	None	8-bit unsigned integer	255	0 -130	1(-65)
Emis_31: Band 31 Emissivity	None	8-bit unsigned integer	0	1 -255	0.0020(+0.49)
Emis_32: Band 32 Emissivity	None	8-bit unsigned integer	0	1 -255	0.0020(+0.49)
the days in clear sky conditions and with valid LSTs	None	8-bit unsigned integer	0	1 -255	NA
the nights in clear sky conditions and with valid LSTs	None	8-bit unsigned integer	0	1 -255	NA

Appendix G: OMNO2 Product Summary

Shortname:	OMNO2
Longname:	OMI/Aura Nitrogen Dioxide (NO2) Total and Tropospheric Column 1-orbit L2 Swath 13x24 km V003
DOI:	10.5067/Aura/OMI/DATA2017
Version:	003
Format:	HDF-EOS5
Spatial Coverage:	-180.0,-90.0,180.0,90.0
Temporal Coverage:	2004-10-01 to 2023-08-12
File Size: 3	5 MB per file
Data Resolution Spatial:	13 km x 24 km
Temporal:	98 minutes